

Constrain the Target, Not the Loss: Action-Space Trust Regions for Critic-Guided Policy Fine-Tuning

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Abstract

Behavior cloning has made visuomotor robot policies easy to train, but their performance is bounded by the demonstrations they imitate. Reinforcement learning offers a way past that bound by letting the policy practice against a learned critic, yet in the regime that matters in practice — a policy cloned from few demonstrations, improved online, rewarded only when the task succeeds — critic-guided fine-tuning is unreliable and frequently destroys the policy it was meant to improve. The cause is that a critic is accurate only on the narrow set of actions it was trained on, while the standard update lets that critic determine both the direction and the magnitude of every policy change. The policy is therefore pushed outside the region the critic understands, where it stops succeeding and, under sparse reward, receives no signal that would guide it back.

We present a fine-tuning rule that separates these two roles: the critic chooses the direction of the update, and the algorithm chooses the distance. Candidate actions are drawn from the frozen pretrained policy, displaced along the critic’s gradient, pulled back whenever they stray beyond a fixed radius of the action they started from, and regressed into a lightweight residual head. The critic, replay data and interaction budget are unchanged from the strongest published residual fine-tuner, so the actor update is the only variable.

Across two benchmarks this yields +9.4 points on the eight hardest LIBERO-90 tasks where backpropagating the critic yields +0.9, and raises robomimic square from 0.382 to 0.91–0.93, while published methods that let the critic set the distance fail to improve their base and in two cases collapse to zero. Ablating our own components isolates why: bounding how far the policy moves per step changes nothing, and the component that matters is the one holding the policy near where it started. The effective constraint is a leash to a fixed anchor, not a speed limit. The update is unchanged across diffusion, flow-matching and one-step bases, and across sparse and dense reward — on dense DMC control it wins outright while the unconstrained update can still fall below its base.

1 Introduction

Behavior cloning is the default way to build visuomotor robot policies, and its ceiling is the demonstrations themselves: teleoperated data is expensive, operators are rarely optimal, and performance saturates as demonstrations accumulate. Reinforcement learning is the natural way past that ceiling, but modern generative policies map noise to multimodal action chunks and expose no tractable likelihood, which rules out PPO-style machinery; the workarounds either change the deployed policy by injecting noise (Zhang et al., 2026) or assume a denoising objective the policy does not have (McAllister et al., 2025). The dominant practical recipe is therefore residual RL: freeze the pretrained policy, learn a small corrective term on top, and score candidates with a learned critic (Sun & Song, 2026).

That recipe has a failure mode we believe is underappreciated, and it is specific to the setting practitioners care about most: a policy cloned from few demonstrations, improved online, with reward that fires only on task success. In that regime, critic-guided fine-tuning frequently makes the policy worse, sometimes catastrophically so. Running three published action-gradient fine-tuners from their authors’ own code, at $125\times$ our interaction budget, none improves its starting policy and two collapse to exactly 0.000 success (Table 2).

The cause is a division of labour nobody chose deliberately. A critic ranks actions well where its data is and is unconstrained — confidently, steeply wrong — beyond it (Fujimoto et al., 2018; 2019; Jain et al., 2025). The standard update nevertheless lets the critic set both the direction of each policy change and, through its gradient magnitude, the distance. Only the first is a question it was trained to answer, and the errors are not symmetric: a wrong direction costs one bad step, while a wrong distance carries the policy out of the region the critic understands — where the further it goes, the better it looks to the critic that sent it.

Figure 1 shows this with nothing hidden. We fit a critic to 25 demonstrations lying on a narrow band of workable actions and follow its gradient from a demonstrated action. The policy walks off the band to a point the critic scores 1.63 — higher than any reward the task can return — where the true reward is exactly 0.000. Repeat with 700 demonstrations and the effect vanishes: the critic has seen enough to know the band. The failure is not a property of critics in general but of post-training from few demonstrations. Sparse reward is what makes it terminal rather than temporary: under dense reward a policy that wanders still receives graded feedback and can recover, and we verify that the same unconstrained update survives there; under success-only reward, a policy that has stopped succeeding sees nothing but zeros and has no signal with which to return.

CAST (Constrained Action-Space Transport) splits the two roles: the critic chooses each action’s direction of change; the algorithm decides how far the policy may end up from where it started. Candidate actions are drawn from the frozen base, displaced along the critic’s gradient, pulled back whenever they stray beyond a fixed radius of their starting action, and regressed into a lightweight residual head. Nothing else in the pipeline changes, so the actor update is the only variable in every comparison we report.

Our contributions:

- A diagnosis of why critic-guided post-training fails from few demonstrations (Section 3.3, Figure 1). We show, on a fitted critic rather than by argument, that the failure is an extrapolation problem with a sharp data threshold: at 25 demonstrations the critic reports a value above anything the task can return at an action whose true reward is zero, and at 700 the effect disappears. We further show that sparse reward is what makes the failure permanent, by exhibiting the same unconstrained update surviving under dense reward.
- An actor update that separates direction from distance (Section 4). Improving actions along $\nabla_a Q$ and regressing the policy onto them is an established family (Li et al., 2024; Psenka et al., 2025; Choi et al., 2026); we adopt that form and add what it lacks — a bound, placed in action space where the harm occurs rather than in the loss, where a penalty must out-shout a gradient of unbounded size and loses (0.549 and 0.601 against bases of 0.573 and 0.610). Two further choices separate our instantiation: targets come from freshly drawn particles off a frozen base rather than buffer actions mutated in place, so stored transitions remain consistent with the dynamics that produced them, and the update needs no likelihood.
- Evidence that the update rule is what wins, under a strict control (Section 5). Holding base, critic, data and budget fixed and varying only the actor update, our rule gains +9.4 points on the eight hardest LIBERO-90 tasks where backpropagating $-Q$ gains +0.9, lifts robomimic square from 0.382 to 0.91–0.93, and transfers unchanged across generator classes. Two controls close the obvious alternative explanations: sweeping each method’s step size an order of magnitude leaves the two in non-overlapping bands, so the gap is not tuning; and deleting only the critic’s gradient direction collapses performance below base, so the gain is not mere caution.

- An ablation that overturns the intuitive explanation (Section 5.4). The natural reading of our own method is that it works by limiting step size. It does not. With the step clip alone success is exactly 0.000 — identical to using no constraint — because a drift assembled from many individually permitted steps arrives just as surely as one large one. What makes the update viable is the term anchored to the frozen base. The effective constraint is a tether to a fixed reference, not a speed limit, and we report this because it contradicts what we expected to find.

2 Related Work

RL fine-tuning of generative robot policies. Policy-gradient fine-tuning of vision-language-action models with sparse task reward is now standard for autoregressive token policies (Li et al., 2025), but transfers poorly to continuous generative heads: likelihood ratios are intractable for diffusion and flow policies, and workarounds either reformulate the denoising chain as an MDP (Ren et al., 2025; Black et al., 2024) — a construction with no analog for one-step generators — or avoid likelihoods entirely (Park et al., 2025; Zhang et al., 2026; Mark et al., 2024). FPO (McAllister et al., 2025) is the strongest likelihood-free ratio — it substitutes differences of conditional flow-matching losses for $\pi_\theta/\pi_{\theta_{\text{old}}}$, without perturbing the deployed policy — but its surrogate requires a denoising loss over a flow time, which a one-step drifting model does not have. The two approaches are complementary: FPO makes the ratio tractable, we make the critic safe to follow. Closest to our setting, DICE-RL (Sun & Song, 2026) fine-tunes a frozen one-step distillation of a flow-matching policy by training a residual head with a backpropagated critic and a filtered BC anchor; it is our primary baseline and the control for our actor-update ablation, since we keep its critic, residual parameterization, and data pipeline unchanged.

Where this update sits. RL for diffusion and flow policies divides into four families (Choi et al., 2026): backpropagating through the denoising chain (Diffusion-QL, and DICE-RL’s residual actor); ratio-based methods that make a likelihood tractable (DPPO, FPO); Q -weighted regression; and action-gradient methods, which improve actions with $\nabla_a Q$ and then regress the policy onto the improved actions. CAST belongs to the last family. Its primitive is not ours: DIPO (Li et al., 2024) updates replay-buffer actions by gradient ascent and fits a diffusion policy to them, and QSM (Psenka et al., 2025) matches the policy’s score to $\nabla_a Q$. What no member of the family supplies (Choi et al., 2026) is a bound on how far the displacement may carry the policy or a restoring force toward a behavior policy — and that gap is the family’s documented weakness: degradation under biased critic gradients, and drift away from the behavior distribution.

Our contribution is precisely that missing piece, and our ablations measure it directly. The displacement is clipped pointwise in action space, a dead-zone anchor restores toward a frozen base, and the update is applied to a residual rather than by overwriting stored transitions (DIPO’s buffer rewrite makes the stored action inconsistent with the transition that produced it). Two measurements tie our system to theirs. Removing the base-restoring anchor reproduces the family’s failure inside our own code exactly — success 0.000, whether or not the pointwise clip is kept (Table 5) — so we can switch their failure on and off. And running DIPO, QSM and DQL from their authors’ implementations on our setting, at $125\times$ our interaction budget, none improves its base and two reach exactly 0.000 (Table 2), which is the same failure observed from the outside. We are careful about the direction of this claim: these methods were developed for dense-reward online control, and the adaptation to sparse-reward manipulation fine-tuning is the one published by Ren et al. (2025), not one we devised. Their difficulty here is evidence about the regime, not a verdict on the methods. Test-time guidance (Zhou et al., 2026) applies the same $\nabla_a Q$ signal without training at all, and is complementary.

Critic-as-compass versus critic-as-judge. That a learned Q -function’s action-gradient cannot be trusted is a recurring diagnosis: overestimation and narrow spurious peaks in TD3 (Fujimoto et al., 2018), extrapolation error in BCQ (Fujimoto et al., 2019), and spurious local optima that make sample-and-select beat gradient ascent (Jain et al., 2025). One re-

sponse treats the critic only as a judge: weighted regression and sample-based improvement (AWR, CRR, MPO, IDQL, QT-Opt) (Peng et al., 2019; Wang et al., 2020; Abdolmaleki et al., 2018; Hansen-Estruch et al., 2023; Kalashnikov et al., 2018) tilt the policy toward $\pi \cdot \exp(Q/\tau)$ using only Q -values. Our method spans this spectrum rather than picking a side: the update form is judge-safe (bounded transport toward targets), while the reward field inside it can still be the analytic gradient where it is trustworthy — and our ablations map exactly when each choice wins. The judge-only methods are also the ones that survive our benchmark: of the published fine-tuners we ran, the one that only weights actions rather than displacing them is the only one to improve its base (Table 2). It pays for that safety by discarding the critic’s most informative output, which is what our bounded update is designed to keep.

Particle transport as a policy update. Regressing a sampler onto displaced particles is the amortized-SVGd idea of Soft Q -Learning (Haarnoja et al., 2017), with kernel repulsion preserving diversity as in SVPG (Liu et al., 2017); drifting models (Deng et al., 2026) make the same operator the pretraining objective of a one-step generator. Drifting Field Policy (Koo et al., 2026) is the closest prior method: it also performs RL on a drifting backbone by drifting fresh policy samples toward critic-ranked top- K candidates. CAST differs in three ways: we update a residual on a frozen base rather than the full policy, we transport with an explicit pointwise trust region and a restoring anchor field (stability contract our toy shows is load-bearing), and our reward field is a dial — including the analytic $\nabla_a Q$ source that dominates when the critic generalizes — rather than a fixed ranking surrogate.

Residual RL and staying on-manifold. Learning a residual on a frozen base is a long-standing recipe for sample-efficient robot RL; DAWN (Ma et al., 2026) identifies the value cold-start and scale pathologies of residual critics that we also observe. The off-manifold drift our toy exhibits is the mechanism behind over-optimization and mode collapse in RL-finetuned generators (Black et al., 2024; Fan et al., 2023; Nakamoto et al., 2023), and forgetting in RL-finetuned language models is likewise governed by divergence from the base (Shenfeld et al., 2025) — which cannot even be measured without likelihoods, reinforcing update rules whose step size is bounded in action space by construction.

In sum, prior work either trusts the critic’s gradient and pays for it off-manifold, or discards it and pays on hard tasks; CAST is, to our knowledge, the first fine-tuner for one-step generative policies whose update form is safe under an arbitrary critic while still admitting the full first-order signal when it is available.

3 Problem Statement

3.1 The setting: post-training a cloned policy

A robot policy has been trained by behavior cloning on tens of demonstrations, and it works, imperfectly. We want it to improve by practising in its environment, with the only feedback being episode success. Three features of this setting drive everything that follows:

- The demonstrations are few. Whatever a learned critic knows, it learned from a thin shell of actions around them.
- The reward is sparse. A policy that stops succeeding receives a stream of zeros, which carries no information about how to succeed again.
- The base policy is worth protecting. It is the only thing in the system that reliably does the task at all.

Formally, states are s , actions are chunks $a \in \mathbb{R}^{H \times d}$ of H control steps flattened to dimension $S = Hd$, and reward is a sparse task-success signal. The pretrained policy maps a state and a noise sample to an action in a single network evaluation,

$$a = \pi_0(s, z), \quad z \sim \mathcal{N}(0, I), \quad (1)$$

so the policy is a deterministic map from noise to action. Following standard residual fine-tuning practice (Ma et al., 2026; Sun & Song, 2026), we freeze it and learn a small residual on top,

$$a = \pi_0(s, z) + r_\theta(s, z), \quad (2)$$

alongside an ensemble critic $Q_\phi(s, z, a)$ trained by temporal difference with a conservative penalty (Kumar et al., 2020). Critic training is identical for every method we compare. The object of study is only the actor update — how the policy consumes the critic it is given.

3.2 Why the obvious tools do not apply

Policy gradients need a likelihood this policy does not have. PPO- and GRPO-style finetuners (Li et al., 2025) require $\log \pi(a | s)$. A one-step generator defines its action distribution implicitly through the noise map, its likelihood is intractable, and unlike multi-step diffusion there is no chain-MDP reformulation to fall back on (Ren et al., 2025; Park et al., 2025). Two routes recover a ratio and neither is free. One changes the deployed policy, adding Gaussian noise to its output so a surrogate density exists (Zhang et al., 2026); we include it as a baseline and it is competitive, but what ships is no longer the policy that was pretrained. The other leaves the policy alone but requires it to be a flow model: FPO (McAllister et al., 2025) builds a clip ratio from differences of conditional flow-matching losses, which needs a denoising loss over a flow time τ that a one-step model does not have. So a critic is what remains, and the question becomes how to consume it.

Every critic-based update hands the critic two jobs. A critic can answer one question well: given this action, which way should it change to score higher. Standard updates ask it a second question at the same time — how far — and take the answer from the magnitude of its gradient. Backpropagating $-Q$ through the actor, as in the DDPG lineage and DICE-RL’s residual actor (Sun & Song, 2026), does this explicitly,

$$\nabla_\theta \mathbb{E}[-Q] = -\nabla_a Q \cdot \partial a / \partial \theta, \quad (3)$$

so a steeper critic produces a larger parameter step. The action-gradient family (Li et al., 2024; Psenka et al., 2025) never differentiates through the policy, but the improved action lands wherever $a + \alpha \nabla_a Q$ falls, so the displacement is again proportional to the critic’s steepness — the same delegation, one level up.

Only the first question is one the critic was trained to answer. It learns to rank actions where it has data; away from that data nothing constrains it, and it can be steep and wrong at once (Fujimoto et al., 2018; 2019; Jain et al., 2025). The two errors then fail asymmetrically. A wrong direction costs a slightly wrong step. A wrong distance costs the data manifold — and once the policy is off it, higher critic scores mean worse behavior, because the critic is scoring a region it has never been corrected on.

One detail decides which remedy can work: the damage is persistent, not sudden. The critic’s extrapolated optimum does not recede as the policy approaches it, so every update points the same way and nothing pulls the policy back. A limit on step size therefore only slows the drift; a force that returns the policy toward its starting point prevents it. Section 5.4 measures both.

3.3 Watching it happen

We make the argument visible in a setting small enough to draw completely: two-dimensional actions, a curved band of workable actions standing in for the data manifold, and a reward that decays away from the band. A critic is fit by regression to 25 demonstrations sampled from the band — only demonstrated actions and their true rewards, exactly what a critic trained on demonstration data sees.

Figure 1(a–c) shows the outcome: the critic is accurate on the band and extrapolates freely beyond it, and following its gradient walks the policy to a point scored $\hat{Q} = 1.63$ — above any achievable reward — where true reward is 0.000, while the bounded update improves the same start ($0.016 \rightarrow 0.040$). The effect has a data threshold, and that threshold defines

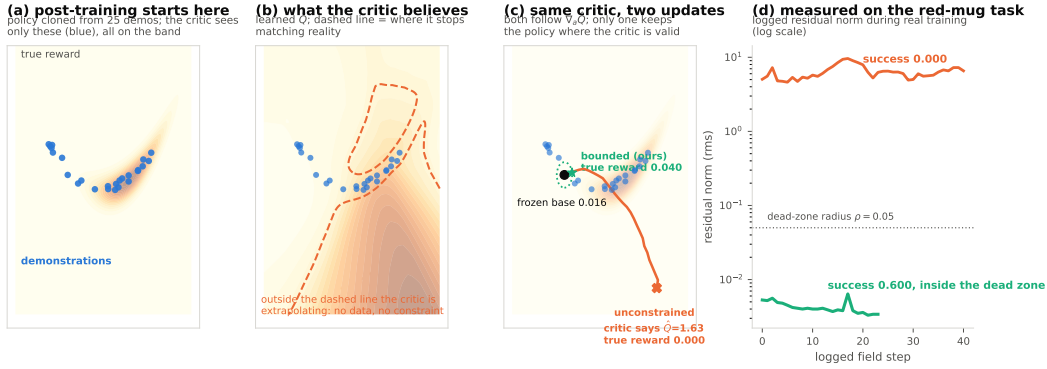


Figure 1: Why critic-guided post-training leaves the data, and what a bound does about it. Panels (a)–(c) are computed, not drawn: a critic is fit by regression to 25 demonstrations, the number a post-training setting actually has. (a) The task’s true reward, and the demonstrations — all of which lie on one narrow band of feasible actions. (b) What the critic learned. On the band it is accurate; beyond the dashed contour it has no data and is extrapolating, and nothing in its training penalises it for being confidently wrong there. (c) Two updates from the same start, both following $\nabla_a \hat{Q}$. The unconstrained one walks off the band to a point its own critic scores $\hat{Q} = 1.63$ — higher than any achievable reward — where the true reward is exactly 0.000. Bounding the displacement around the frozen base keeps the policy where the critic is valid and improves it ($0.016 \rightarrow 0.040$, $2.5\times$). The gap between what the critic claims and what the policy earns is the whole failure. (d) The same mechanism in the real system: the residual norm logged during LIBERO red-mug training, from the runs behind Table 5. Without a base-restoring term the residual immediately sits two orders of magnitude outside the dead zone and success is 0.000; with it the residual stays an order of magnitude inside the dead-zone radius — the bound is slack, not binding — and success is 0.600. The regime matters: repeating (a)–(c) with 700 demonstrations instead of 25 removes the effect entirely (critic then learns the band and its gradient is trustworthy), which is why this paper scopes its claims to post-training from few demonstrations.

this paper’s scope: at 200 demonstrations the critic’s endpoint earns 0.901 against its own estimate of 0.910, so with enough data the gradient is worth following without any bound. Critic gradients are not untrustworthy in general — only in the few-demonstration regime where post-training happens, and every claim in this paper is scoped to it.

The same boundary can be crossed a second way, which rules out the objection that our two-dimensional picture is a special case. Holding the amount of data fixed and raising the dimension of the action space shrinks the fraction of that space the critic has seen, and produces the identical two regimes (Figure 2): with 2-D actions the critic’s gradient generalizes and backpropagation is the best available update, while with 16-D actions and the same critic budget the same update drives true reward to zero as its internal Q -estimate explodes. Coverage is the variable in both experiments; dimension and demonstration count are two ways of changing it.

4 Method

Section 3.3 left us with a concrete requirement: follow the critic where it has data, and prevent the policy from wandering to where it does not. Crucially, this is a requirement about where the policy ends up, not about how fast it travels — a distinction the experiments turn out to hinge on.

Where the critic stops being reliable is not something we can compute — if we could estimate its competence region we would already have solved the problem — so we approximate that boundary by the only landmark available: the frozen policy the critic was trained around.

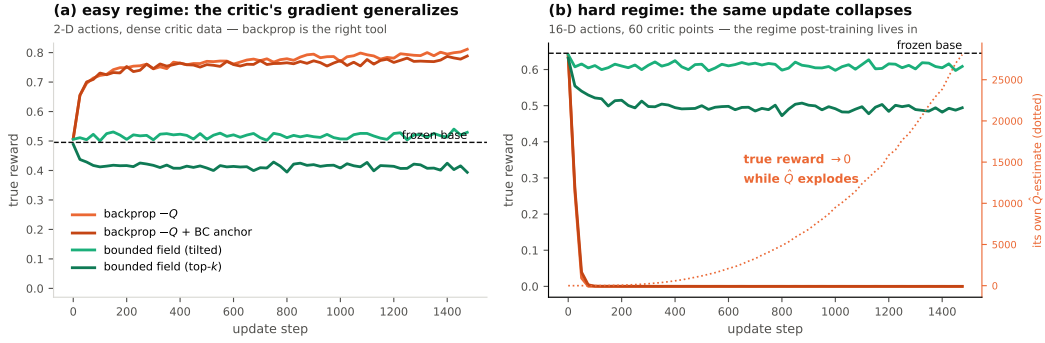


Figure 2: The two-regime toy (3-seed means). (a) Easy regime (2-D actions, dense critic data): the critic’s gradient generalizes and backpropagating $-Q$ — with or without a BC anchor — is the best update, clearing both the base and the bounded field update. (b) Hard regime (16-D actions, 60 critic points, the coverage post-training actually has): the same updates drive true reward to zero within the first hundred steps, while the collapsing arm’s own $\hat{Q}$ -estimate grows without bound (dotted, right axis) — the critic rates the policy ever higher as it gets worse, and the BC anchor does not save it. The bounded field updates hold near the base throughout. A useful fine-tuner must exploit the gradient where it is trustworthy and survive where it is not. This sweeps the dimension of the action space at fixed data, and Figure 1 sweeps the amount of data at fixed dimension; both cross the same boundary, which is what the critic’s coverage of the region the update moves into.

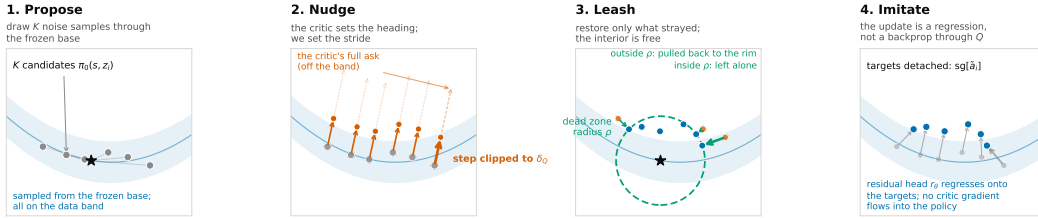


Figure 3: CAST in four steps (schematic; one candidate drawn bold and followed through the pipeline). (1) Propose: the frozen base maps K noise samples to candidate actions on the band the demonstrations support. (2) Nudge: the critic’s raw ask points far off that band (dashed); CAST keeps only its direction and steps at most δ_Q (solid). (3) Leash: candidates that end up outside the dead zone of radius ρ around the base action are pulled back to its rim (green); those inside are left alone — the interior is free. (4) Imitate: the corrected targets are detached ($\text{sg}[\cdot]$) and the residual head is regressed onto them; no critic gradient ever backpropagates into the policy. The base is any frozen noise-to-action map.

Actions near its output the critic has seen; actions far from it, it has not. The constraint therefore takes the form of a radius about the base policy’s own action.

CAST — Constrained Action-Space Transport — turns this into an update rule. It transports candidate actions toward higher critic value, and it constrains that transport in action space, on the regression target, before anything is fitted. The transport primitive itself (move samples along a velocity field, regress the network onto the moved samples) is inherited from the action-gradient family (Li et al., 2024; Psenka et al., 2025); the contribution is the constraint. Nothing in the update needs a likelihood, a denoising objective, or the generator’s internals — only a frozen noise-to-action map, K samples from it, and a critic — so it applies unchanged across one-step, flow-matching and diffusion policies (Section 5.8).

Table 1: CAST’s parts, and how each is designed. One row per component of Equation (4). Defaults are the family-shared recipe (LIBERO); robomimic differs only where noted. Each part’s sensitivity is measured in the referenced experiment.

component	role	design	default	ablated in
candidates K	coverage of the base’s action modes	noise samples through the frozen base + residual	16	Table 7
reward field V_Q	direction of improvement	normalized critic gradient $\eta_Q \nabla_a \bar{Q} / q_{scale}$	$\eta_Q=1$ (0.5 robomimic)	Table 11
step clip δ_Q, δ_{tot}	stride: caps each displacement	rescale to norm at most δ	$\delta_{tot}=0.15$	Table 5
anchor V_{anc}	keeps targets near the base	dead-zone restoring field, Equation (8)	$\eta_{anc}=1$	Table 5
dead zone ρ	region where the critic is trusted freely	radius about the base action ($\rho=0$ = always-on pull)	0.05 rms (LIBERO); 0 (robomimic, DMC)	Table 9
regression (Equation (4))	converts targets into parameters	MSE onto detached targets; no critic gradient	—	Table 3

4.1 CAST in four steps

The critic’s answer is treated as advice about direction; how far to act on it is our choice, and the critic is never differentiated through. One update does four things:

1. Propose. Draw K noise samples and push them through the frozen base plus the current residual, giving K candidate actions for this state.
2. Nudge. Ask the critic which direction improves each candidate, and take a step of bounded length in that direction — the critic sets the heading, we set the stride.
3. Leash. If a candidate has drifted more than ρ away from what the frozen policy proposed, pull it back toward that action. Inside ρ nothing happens, so the critic has free rein where it is trustworthy.
4. Imitate. Detach the corrected candidates and train the residual head by regression onto them. They are data now, not a computation graph.

Two consequences are worth noticing immediately. Because step 4 is plain regression onto fixed targets, the size of the parameter update is set by us, not by how steep the critic happens to be. And because step 3 measures distance from the frozen policy rather than from the previous iterate, the reference point never moves — which, as Section 5.4 shows, is exactly what separates a constraint that works from one that does not.

Concretely: for each state in the batch we draw K noise samples and form the current action candidates $a_i = \pi_0(s, z_i) + r_\theta(s, z_i)$, treated as constants (detached). Each is displaced by a combined velocity,

$$\tilde{r}_i = \text{sg} \left[r_i + \text{clip}_{\delta_Q} (V_Q(s, z_i, a_i)) + V_{anc}(r_i) \right], \quad \mathcal{L}_{actor} = \frac{1}{K} \sum_i \|r_\theta(s, z_i) - \tilde{r}_i\|^2, \quad (4)$$

where $\text{sg}[\cdot]$ is stop-gradient and $\text{clip}_\delta(v) = v \cdot \min(1, \delta/\|v\|)$ rescales a field vector to norm at most δ .

Matching this to the four steps: V_Q is the critic’s direction (step 2), clipped to length δ_Q ; V_{anc} is the leash (step 3), silent while $\|r_i\|$ stays within ρ ; the total displacement is clipped to δ_{tot} ; and $\text{sg}[\cdot]$ makes the result a fixed regression target (step 4). The critic is trained separately (Section 4.4). Table 1 collects every design choice, its role, and the value we run.

Why displace targets rather than differentiate. Writing the update as regression onto displaced targets is what makes the boundary enforceable at all. The quantity we need to bound — how far the policy’s action ends up from the base policy’s action — is a property of the target, and here it is available explicitly, before any gradient is taken: we can inspect each displaced target, compare it to the action it came from, and correct it. Backpropagating $-Q$ offers no such handle, because the distance travelled is an emergent consequence of a parameter step and is not represented anywhere the algorithm can act on. Two secondary properties follow from the same choice. The regression target is detached, so the critic never differentiates through the actor and cannot inject unbounded curvature into the parameter update. And because targets are formed from freshly drawn samples rather than from stored actions rewritten in place, replayed transitions continue to describe the dynamics that produced them — a consistency DIPO’s buffer rewrite gives up (Li et al., 2024).

What the clip does and does not guarantee. The clip bounds target transport, and only that:

$$\|\tilde{r}_i - r_i\| \leq \delta_{\text{tot}} \quad \text{for every particle } i. \quad (5)$$

It is tempting to read this as a bound on how far the policy moves, and an earlier version of this work did. That inference is invalid. The targets are amortized through a shared network by a gradient step, so to first order the realized change at particle i is

$$\Delta r_i \approx \alpha \sum_j J_i J_j^\top (\tilde{r}_j - r_j), \quad J_i = \partial r_\theta(s, z_i) / \partial \theta, \quad (6)$$

and the Jacobian coupling across particles carries no guarantee that $\|\Delta r_i\| \leq \delta_{\text{tot}}$. This is precisely the objection we raise against parameter-space gradient clipping (Section 3.2), and it applies to target-space clipping too: bounding what the network is asked for does not bound what it does. A per-particle guarantee would require an output-space proximal step, exact interpolation, or a tabular actor — none of which we impose.

The distinction matters, but it does not undermine the construction, because a per-particle guarantee is not what the update needs. A drift assembled from many individually permitted steps is still a drift, since the critic keeps pointing the same way and nothing returns the policy — which is why the constraint that matters is one whose strength grows with the displacement already accumulated, rather than one that merely rate-limits each step. What we establish empirically is narrower and sufficient: with no action-space bound of any kind the update destroys the policy, and with either bound present it does not (Section 5.4).

4.2 The reward field V_Q

The default (and headline) reward field is the normalized critic gradient evaluated at each particle,

$$V_Q^{\text{grad}}(s, z, a) = \eta_Q \frac{\nabla_a \bar{Q}_\phi(s, z, a)}{q_{\text{scale}}}, \quad q_{\text{scale}} = \mathbb{E}[|\bar{Q}_\phi|], \quad (7)$$

where $\bar{Q}_\phi$ is the conservative ensemble aggregate. This is deliberately the same scalar the DICE-RL actor optimizes: in the small-step, unclipped limit, minimizing Equation (4) recovers the backpropagated update direction exactly (Proposition 1), so any performance difference is attributable to the update form — the anchored target regression — and not to a different reward signal.

The kernel transport operator. The lower-resolution reward fields below and the distributional anchor variant share one particle transport operator, which we take off the shelf (Deng et al., 2026); it is a convenience for turning a set of target particles into a velocity field, and nothing in the method depends on its particular form. Given query particles x , attractor particles p^+ (optionally weighted), and repulsor particles p^- , distances are normalized by a global scale and, for each kernel radius R in a small set, a soft mutual-matching affinity $A = \sqrt{\text{softmax}_{\text{row}}(-d/R) \odot \text{softmax}_{\text{col}}(-d/R)}$ converts matched attractors into pulls and matched repulsors into pushes; forces are summed over radii to give $\mathcal{V}(x \mid p^+, p^-)$ (full form in Appendix A). Intuitively, $\mathcal{V}$ moves a particle cloud so its density flows toward p^+ and away from p^- — it needs only samples of the target, never a density or a gradient.

Because V_Q enters Equation (4) only as a velocity, the critic can be consumed at lower resolution when its gradients are not trustworthy, using only Q -values:

- zeroth-order (top- k): rank the K particles by Q and set $V_Q = \mathcal{V}(a \mid \text{top-}k \text{ particles, all particles})$ — the pretraining kernel transport toward the critic’s preferred candidates;
- tilted: let every particle attract with weight $K \cdot \text{softmax}_i(\text{adv}_i/\tau)$, transporting the cloud toward its own $\exp(Q/\tau)$ -tilted density, the canonical improvement target of Peng et al. (2019); Abdolmaleki et al. (2018).

These are not competing methods but settings of a dial — how much of the critic’s local structure the update trusts: full first-order information, ranking only, or soft value weights.

Section 5 maps the regimes: the gradient source dominates wherever the critic generalizes, while the value-only sources close the gap precisely in the weak-critic regime predicted by Section 3.3.

4.3 The anchor field

The kernel reward fields vanish beyond their largest radius and the gradient field carries no notion of the data manifold, so Equation (4) alone would let the residual random-walk. The anchor is a restoring field with a dead zone,

$$V_{\text{anc}}(r) = -\eta_{\text{anc}} \left(1 - \frac{\rho}{\|r\|}\right)_+ r, \quad (8)$$

which is inert inside the trust radius ρ and pulls back only the excess residual norm beyond it. Equivalently, Equation (8) is the negative gradient of a hinged radial potential,

$$U(r) = \frac{1}{2} \eta_{\text{anc}} (\|r\| - \rho)_+^2, \quad V_{\text{anc}}(r) = -\nabla_r U(r), \quad (9)$$

so it removes exactly the radial excess and nothing else. Setting $\rho = 0$ gives $U(r) = \frac{1}{2} \eta_{\text{anc}} \|r\|^2$ — the quadratic residual regularizer of the backpropagated control — so the dead zone is a hinge generalization of that penalty, with ρ interpolating between them. Unlike a $\lambda \|r\|^2$ penalty — which competes with an unbounded gradient and loses (Section 3.3) — Equation (8) composes with the clipped reward field as a velocity, so the residual scale is set geometrically rather than by a loss-weight arms race. Heuristically, if the reward field were stationary, radially outward and saturated at $\|V_Q\| = \delta_Q$, balancing it against the anchor outside the dead zone gives $\delta_Q = \eta_{\text{anc}} (\|r\| - \rho)$, i.e. $\|r\| = \rho + \delta_Q / \eta_{\text{anc}}$. We use this only as a scale estimate when choosing η_{anc} : a learned critic’s field varies in direction and magnitude with r , so there may be several fixed points, cycles, or none, and we claim no unique equilibrium.

4.4 Relation to the backpropagated actor; critic

Proposition 1 (Small-step equivalence). At the current parameters, the gradient of Equation (4) is $\nabla_{\theta} \mathcal{L}_{\text{actor}} \propto -(\text{clip}_{\delta_Q}(V_Q) + V_{\text{anc}})^\top \partial r_{\theta} / \partial \theta$. With $V_Q = V_Q^{\text{grad}}$, no clipping ($\delta_Q, \delta_{\text{tot}} \rightarrow \infty$) and $\rho = 0$, this is the DICE-RL residual actor gradient with anchor weight η_{anc} .

CAST is therefore a strict generalization of its control: the new degrees of freedom are the pointwise action-space clip, the dead-zone anchor, and the resolution dial on V_Q — each of which we ablate.

Remark (not just clipped gradient ascent). One might object that Equation (4) with V_Q^{grad} is “DDPG with clipping.” The distinction is where the bound lives. Clipping ∇_{θ} bounds a parameter step, whose effect on the policy output is unknown and state-dependent; Equation (4) bounds the policy’s motion at every particle, which is the quantity that off-manifold failure depends on (Section 3.3). Moreover, only the regression form makes the two other components expressible at all: a dead-zone anchor and a value-only V_Q have no loss-gradient counterpart. The ablations in Section 5.6 test each distinction in isolation. The critic is unchanged from DICE-RL: ensemble TD with a CQL-style penalty pushing Q down at policy actions and up at data actions; field construction never leaks gradients into the critic (gradients w.r.t. actions only). Algorithm 1 summarizes one joint update.

5 Experiments

We test four claims. The first asks how the method compares to what a practitioner could use today. The second is the causal claim, and is the paper’s main result. The third and fourth ask which part of the method is responsible for it.

- H1 CAST matches the strongest existing fine-tuners. The methods that, like it, let the critic displace actions — but with no action-space constraint — do not improve their base at all.

Algorithm 1 CAST: one update (V_Q^{grad} variant)

Require: frozen base π_0 ; residual r_θ ; critic Q_ϕ ; steps $\eta_Q, \eta_{\text{anc}}$; clips $\delta_Q, \delta_{\text{tot}}$; dead zone ρ ; particles K

- 1: sample batch of states s ; draw $z_i \sim \mathcal{N}(0, I)$, $a_i \leftarrow \pi_0(s, z_i) + r_\theta(s, z_i)$ $\triangleright$ no grad
- 2: $V_i \leftarrow \text{clip}_{\delta_Q}(\eta_Q \nabla_a \bar{Q}_\phi(s, z_i, a_i) / q_{\text{scale}})$ $\triangleright$ or top- k / tilted kernel field
- 3: $V_i \leftarrow V_i - \eta_{\text{anc}}(1 - \rho / \|r_i\|)_+ r_i$ $\triangleright$ dead-zone anchor, Equation (8)
- 4: $\tilde{r}_i \leftarrow \text{sg}[r_i + \text{clip}_{\delta_{\text{tot}}}(V_i)]$ $\triangleright$ policy moves $\leq \delta_{\text{tot}}$, always
- 5: $\theta \leftarrow \theta - \alpha \nabla_\theta \frac{1}{K} \sum_i \|r_\theta(s, z_i) - \tilde{r}_i\|^2$
- 6: update ϕ : ensemble TD + CQL penalty $\triangleright$ unchanged from Sun & Song (2026)

H2 Change only the actor update, and CAST gets more out of the same critic than backpropagating $-Q$ does.

H3 Take the action-space constraint away, and the update destroys the policy.

H4 The constraint has to act on the action target. The same shape placed in the loss, or on the parameter gradient, does not work.

Two supporting analyses follow: how much of the critic the update should consume (Section 5.7), and whether the result depends on the generator class the update is attached to (Section 5.8).

Throughout, “—” means not measured. No cell is inferred from a related run.

5.1 Setup

Benchmarks. Two, chosen because their critics differ. robomimic can and square are state-based, single-task, and densely covered, so the critic generalizes well; we report 300-episode evaluations averaged over the last three checkpoints, on 3 seeds. The eight hardest tasks of LIBERO-90 are image-based and sparsely rewarded, so the critic is far weaker; we report powered evaluation — 100 rollouts $\times$ 3 evaluation seeds. Both run inside the official DICE-RL harness with only the actor update replaced, so critic, replay, demonstrations and evaluation are byte-identical to the baseline.

How we report success. Fine-tuners in this literature start from different pretrained policies, so absolute success confounds the fine-tuner with its base. Where methods do not share a base we give each method’s own base next to its result, and compare $\Delta = \text{fine-tuned} - \text{its own base}$. Only in Section 5.3, where every row is literally the same checkpoint, do we compare absolute numbers.

Methods. Every method on the drifting base shares the residual parameterization, critic, data and budget, so only the actor update varies.

- CAST — Equation (4) with the gradient reward field V_Q^{grad} .
- Backprop actor — the DICE-RL residual actor (Sun & Song, 2026): backpropagated $-Q$ with a filtered behaviour-cloning penalty, run on our drifting base. It reads the identical critic through the identical residual head, so any gap is attributable to the update form. Note that it already carries two constraints of its own: the BC penalty in its loss, and gradient-norm clipping at 1.0 on the actor. Both matter in Section 5.6.
- FM + DICE-RL — the published pipeline, unmodified, on its own flow-matching base.
- Classic diffusion-RL fine-tuners (DIPO, QSM, DQL, IDQL, AWR) — official implementations, unmodified, from their own diffusion base at their own published budget.
- CAST (top- k) and (tilted) — the same update with the reward field built from the critic’s rankings or its soft values instead of its gradient (Section 5.7).

Table 2: Fine-tuners on robomimic square. Classic baselines run the DPPO authors’ adapted configurations verbatim at their published budgets ($\sim 125\times$ our interaction); the lower block pairs DICE-RL with CAST on one shared base per policy class. 300-episode evaluations. Markers: *stricter success rule, Δ vs. its 0.348 reading of this base (see text); $^\circ$ still training; ¶ shared retrained base (0.600); †† matched 496K, see text for the $1.6\times$ speed result; z zeroth-order variant; ¶ flow-matching base 0.570; §§ converged 0.887 on the original-architecture base, see text.

method	policy class	own base	fine-tuned	Δ
DIPO (Li et al., 2024)	diffusion	0.600*	0.156	−0.444
QSM (Psenka et al., 2025)	diffusion	0.600*	0.000	−0.600
DQL	diffusion	0.600*	0.000	−0.600
IDQL	diffusion	0.600*	0.327	−0.273
AWR	generic	0.600*	0.393	−0.207
DPPO (Ren et al., 2025)	diffusion	0.600	0.678 $^\circ\S\S$	+0.078
adding CAST to DICE-RL, paired within each base				
diffusion + DICE-RL (Sun & Song, 2026) ¶	diffusion	0.600	0.965 ††	+0.365
diffusion + DICE-RL + CAST ¶	diffusion	0.600	0.959 †† , $1.6\times$ faster	+0.359
drifting + DICE-RL	one-step drifting	0.382	0.880	+0.498
drifting + DICE-RL + CAST	one-step drifting	0.382	0.924	+ 0.542
flow matching + DICE-RL (Sun & Song, 2026)	flow matching	0.570	0.831 ¶	+0.261
flow matching + DICE-RL + CAST z	flow matching	0.570	0.934 ¶	+ 0.364

Hyperparameters are shared across CAST and its variants, one recipe per benchmark and never per task: $\delta_{\text{tot}}=0.15$, $\eta_{\text{anc}}=1.0$, $\tau=1$, $k=4$, with reward step $\eta_Q=0.5$ on robomimic and $\eta_Q=1.0$ on LIBERO, and a dead-zone radius $\rho=0.05$ in dimension-independent rms units on LIBERO.

5.2 H1: where the method stands among existing fine-tuners

Claim. CAST matches the strongest existing fine-tuners, and the methods closest to it — those with no action-space constraint — do not improve their base at all.

Verdict. Both halves hold: on the retrained diffusion base CAST reaches DICE-RL’s ceiling at less compute (0.955 against 0.941 at 352K, converging by 496K), it matches the strongest published pipeline from the drifting base, and every unconstrained-critic method sits at or below its base.

Against the published flow-matching pipeline, the drifting-base CAST reaches parity (0.912–0.930 against 0.923–0.934, from a base of 0.382), and the diffusion-base pair passes it (both arms saturate near 0.96).

The DPPO cell (0.678 $^\circ$) is quoted on the shared 0.600 base and is still climbing on its $\sim 12\text{M}$ -step schedule; the same method run to convergence on DPPO’s original-architecture diffusion base reaches 0.887 (11.7M steps), which we note here rather than in the table because it is a different base and not directly comparable to the shared-base rows.

Two caption markers deserve their numbers here. The starred baselines’ harness scores success more strictly — a full action chunk inside the success state — and reads the shared checkpoint at 0.348; their Δ is computed against that reading, so up to 0.252 of a starred drop is the measurement rule rather than degradation. And the paired rows tie at the environment’s ≈ 0.96 ceiling because nothing higher exists to win: the separation is speed, with CAST first reaching 0.90 at 128/160/160K steps against DICE-RL’s 224/240/256K — its slowest seed beats DICE-RL’s fastest, $1.6\times$ less interaction on average.

The collapse rows above the line are the ones that matter for this paper, and they have now been measured twice: QSM and DQL fall to exactly 0.000 from the retrained 0.600 base just as they did from the weak base it replaced. A base twice as strong changes nothing about the failure.

We checked our own setup before reporting a collapse. Every hyperparameter matches the published configuration ($n_{\text{critic warmup}}=5$, actor/critic learning rates $10^{-5}/10^{-3}$, q_{grad} coefficient 10, $\tau=0.005$, buffer 5000, replay ratio 16, batch 1000), and it gets more training than published, not less (300 against 201 iterations). The failure is immediate rather than gradual: base level (0.284) at iteration 0, zero by iteration 10, unchanged for the next 290. Both losses converge (actor 0.002, critic 0.01) rather than diverging. That is the signature of a degenerate fixed point — once the policy stops succeeding every reward is zero, the critic correctly learns $Q \approx 0$ everywhere, and $\nabla_a Q$ carries no signal with which to recover. An earlier draft caveated that our then-weaker base (0.279) might explain the collapse; the retrained 0.600 base has since reproduced it exactly, so that caveat is retired.

The five rows split cleanly, and the split is the point. Group them by what the critic is allowed to do. DIPO, QSM and DQL let the critic displace actions — through an action-space gradient step, a score matched to $\nabla_a Q$, or backpropagation through the policy — with nothing bounding the displacement. All three fail outright: two at exactly 0.000, one at +0.004 after 24M environment steps. AWR lets the critic only weight actions the policy already produced, and it is the one method that improves (0.407, +0.128): with no displacement there is nothing for a missing bound to break. IDQL sits between — it never displaces either, but its Q-based action selection still exploits critic error, and it ends below its base (0.041). The failure is therefore not “critic-guided fine-tuning is unstable,” and not our harness: it tracks precisely whether the critic controls where actions move. That is the quantity our constraint bounds. The displacing methods are the unconstrained corner of our own design space, measured externally in their authors’ code at $125\times$ our interaction — and Section 5.4 reproduces exactly their behaviour inside our system by switching our constraint off. What separates CAST from that family is not the signal, the budget, or the policy class; it is the bound. AWR’s positive result sharpens rather than weakens the claim: consuming the critic safely was never the hard part — weighting schemes have always done so — but they leave the critic’s most informative output, its action gradient, on the table (Section 5.7 measures what that costs on the harder task).

CAST as a drop-in on an existing fine-tuner. The rows above compare whole systems, which leaves open whether CAST’s advantage is the update rule or the rest of our pipeline. The cleanest test is to change nothing except the update. Taking DICE-RL on the same pretrained diffusion base that every classic baseline uses, and switching only its actor from the residual update to CAST’s field-target update, raises 300-episode success from 0.861 to 0.937 — +7.6 points, from a one-line configuration change, on top of a method that was already working well. This is the sense in which CAST is a technique rather than a system: it is additive to an existing fine-tuner, not a replacement for one. The corresponding cell on a flow-matching base is not yet measured at matched base and budget, so we do not report it.

We tried to rescue the collapsing baselines, and could not. Reporting a method at settings that make it fail invites the objection that it is merely mis-tuned. So we re-tuned. For QSM we swept the score coefficient, for DQL the Q -loss weight η , for DIPO the action learning rate — in each case toward the value that would match our critic’s gradient scale. All three ended at 0.000: QSM and DIPO collapse from the base within one evaluation and stay there, DQL flickers back once and then dies (this sweep predates the base retrain; the collapse itself replicates on the 0.600 base).

The reason is the paper’s point in miniature. Two of the three knobs are loss-scale constants, and Adam is invariant to loss scale: a $6.5\times$ change in QSM’s coefficient moved the policy displacement from 0.030 to 0.033. The one knob that acts on actions directly — DIPO’s action learning rate — destroyed a baseline that had been working at 0.283. Constants in the loss do not control how far the policy moves in action space, and the one constant that does is a blunt instrument; what these methods lack is a bound on the displacement itself.

Table 3: Matched actor-update comparison on robomimic (300-episode evaluations, last-3-checkpoint average; s42–s44 are independent training seeds from one base per task). Only the actor update differs between the two RL rows. [†]ended before the matched 20K budget.

actor update	how the critic is used	can			square		
		s42	s43	s44	s42	s43	s44
frozen base, no RL	critic never queried		0.877			0.382	
DICE-RL actor (baseline)	Q differentiated through the policy	0.970 [†]	0.946 [†]	0.940	0.870	0.909	0.861
CAST actor (ours)	Q only queried; targets displaced, bounded, regressed	0.988	0.993	0.978	0.912	0.930	0.929

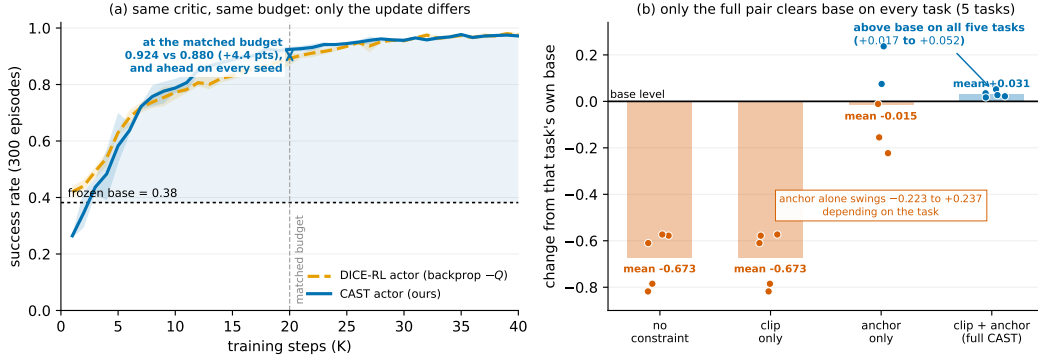


Figure 4: Why CAST wins, and what the full factorial says about its guards. (a) robomimic square (drifting base), 300-episode evaluations; line is the seed mean, band the min–max over three seeds. Both arms share the base, critic, data, replay and budget — only the actor update differs. At the matched 20K budget CAST reaches 0.924 against the backpropagated actor’s 0.880, and is ahead on every seed; both converge near 40K, so the gain is in what the update extracts per step, not in the asymptote. (b) the constraint factorial over all five tasks of Table 5, plotted as change from each task’s own base (bars: means; dots: tasks). Removing the anchor is fatal with or without the step clip — every such run lands at 0.000 success, a mean change of -0.673 — which is the paper’s central negative result: many individually capped steps still compound into a walk off the data. The anchor alone survives but swings from $+0.237$ to -0.223 depending on the task (mean -0.015). Full CAST is the only arm above base on every task ($+0.017$ to $+0.052$). The anchor is necessary; the clip is what makes the result uniform.

5.3 H2: the update form is what converts critic signal into success

Claim. Change only the actor update, and CAST gets more out of the same critic than backpropagating $-Q$ does.

Verdict. Holds on both benchmarks, on every seed, on both bases. This is the paper’s main result.

A third robomimic task, transport, is reported in Appendix C rather than here: it is the setting where the bound must cost — a near-zero-success base needing corrections larger than the trust region, under a critic made trustworthy by abundant successful experience — and the converged numbers (0.594 for CAST against 0.912 for the backpropagated actor at a matched 2.24M steps) are the measured form of that prediction, discussed there in full.

Can is close to saturated and mainly checks that nothing breaks: CAST reaches 0.978–0.993 from a base of 0.877, the backprop actor 0.940–0.970. Square is the discriminative task. Its base solves only 0.382 of episodes; CAST adds 55 points to reach 0.91–0.93 on every seed, where the backprop actor reaches 0.86–0.91. Same critic, same data, same budget, different update: the field-target form extracts more, on every seed of both tasks.

Figure 4 shows the difference is in speed as much as in endpoint. CAST reaches 0.91 on every seed by 16–18K steps; the backprop actor gets there at 19–21K on two seeds and

Table 4: LIBERO hard-8, per-task fine-tuning (task-ID routing). Powered evaluation: 100 rollouts $\times$ 3 seeds; one shared recipe across all tasks (Table 11). Bold: best fine-tuner per row, within-block only. $\ddagger$ provisional GRPO protocol (see text); “—” unrun.

task	drifting base				flow-matching base				
	base	backprop	CAST $_{\eta_1}$	CAST $_{\eta_2}$	base	DICE	GRPO	CAST $_{\eta_1}$	CAST $_{\eta_2}$
Kitchen-1: open drawer, put bowl in	0.818	0.820	0.831	0.857	0.897	0.873	0.800 ‡	0.900	0.887
Kitchen-3: turn on stove, put pan on	0.690	0.703	0.737	0.752	0.700	0.730	0.700 ‡	0.690	0.727
Kitchen-5: ketchup into top drawer	0.610	0.587	0.694	0.729	0.690	0.682	0.700	0.738	0.773
Living-2: orange juice into basket	0.785	0.800	0.842	0.853	0.917	0.915	0.897	0.897	0.913
Living-5: red mug onto plate	0.573	0.550	0.826	0.812	0.787	0.809	0.883	0.878	0.887
Study-1: book into front caddy slot	0.648	0.663	0.710	0.752	0.683	0.690	0.600 ‡	0.667	0.783
Study-1: book into right caddy slot	0.578	0.583	0.657	0.648	0.620	0.592	0.603	0.608	0.563
Study-3: book into front caddy slot	0.588	0.653	0.607	0.635	0.733	0.708	0.550 ‡	0.670	0.680
mean over the eight tasks	0.661	0.670	0.738	0.755	0.753	0.750	0.717 ‡	0.756	0.777

not at all within the window on the third, trails at the matched budget on every seed, and catches up only near 40K where both converge to 0.97–0.99. A bounded update buys a faster, never-worse path to the same asymptote.

Where the ablations live. CAST has four moving parts — a step clip δ_Q , a dead-zone anchor of radius ρ , the choice of how the critic scores candidates, and the target-regression form itself — and each is taken apart separately later in this section, so we list them once here rather than interleaving them with the headline result. Table 5 switches the clip and the anchor on and off independently (the full 2×2); Table 9 sweeps the dead-zone radius ρ from 0 to ∞ ; Table 7 moves the same constraint between the loss, the parameters and the action target, including a hard projection; Table 7 replaces the critic’s gradient with coarser signals (top- k , tilted); and Table 9 is a leave-one-out over the whole LIBERO recipe. Table 8 is not an ablation but a scope test on a different benchmark family.

LIBERO hard-8. Here we deploy CAST as a routing system: one frozen multitask drifting base, one small per-task residual head and critic fine-tuned on each task’s own interactions, and task-ID routing at deployment. This is the architecture every published LIBERO RL fine-tuner effectively uses. One recipe is shared across all eight tasks with no per-task tuning.

The verdict spans all eight tasks. On the drifting base CAST gains +3.9 to +23.9 points per task (mean +9.4 at $\eta_Q=2$, +7.7 at $\eta_Q=1$) and improves on the base for every one. The backprop actor, reading the same per-task critics, averages +0.9 and drops below base on the two highest-headroom tasks. The robomimic ordering transfers intact to the harder benchmark.

The same recipe on a flow-matching base is a wash. With all eight cells now measured, CAST on the flow-matching base averages 0.756 against that base’s 0.753: +0.3 points, three tasks up and five down. The two tasks it does improve are the two with real headroom — the ketchup task (+4.8) and the red-mug task (+9.1), both of which the drifting base also found hard — while on tasks the base already solves well it costs a little, most visibly −6.3 on the Study-3 caddy task. The flow-matching pipeline’s competing fine-tuners behave the same way: DICE-RL and GRPO also leave that base essentially where they found it.

We report this rather than restricting the claim to the base that flatters it. Two readings are consistent with the evidence and we cannot yet separate them. The first is headroom: the flow-matching base starts 9 points higher (0.753 vs 0.661) and is closer to what this critic and this budget can support, so there is less to win and more to break. The second is that the drifting base’s one-step noise-to-action map is unusually well matched to an update that displaces sampled actions, which would make part of our robomimic and LIBERO margin a property of the base rather than of the update alone. The drop-in result on a diffusion base (Table 2: 0.861 $\rightarrow$ 0.937) argues against the strong form of the second reading, since that base is neither drifting nor one-step. What we can state without qualification is the

conditional: where the critic carries signal and the base leaves room, the update form is what converts one into the other.

Two implementation details were load-bearing and are ablated in Section 5.10. The dead zone must be measured in dimension-independent rms units: in vector-norm units the base residual already exceeds ρ at LIBERO’s $S=112$ chunk dimension, so the anchor is active from the first update and freezes the residual. And the weaker critic gradient here rewards a doubled reward step, which the constraint makes safe to take: $\eta_Q=2$ lifts the mean from 0.738 to 0.755 and helps on six of eight tasks, with no setting on any task falling below its base.

Comparison to a critic-free policy gradient. The four marked GRPO cells in Table 4 are provisional — shorter runs scored by a single 20-rollout evaluation rather than the powered protocol — so the GRPO row mean mixes protocols; over its four matched cells it is 0.771. On the one flow-matching task with real headroom, GRPO and CAST extract the same amount (0.883 and 0.878 on the red-mug task, both about +9.5) while the backprop actor manages +2.2. On the ketchup task CAST leads (0.738 against 0.700). On the tasks where nothing gains, all three sit together. So the separation on this benchmark is not CAST against the world; it is update form against backpropagation-through-the-critic. CAST matches what a policy-gradient method extracts without needing the noise-injection surrogate that makes likelihood ratios definable for a one-step generator in the first place.

Where H2 stops: the shared-critic control. The alternative to routing is one policy and one critic trained on all eight tasks jointly. This flips the critic’s situation — our diagnostics show the shared critic’s action-gradient collapsing on most tasks, with per-task $\|\nabla_a Q\|$ spanning 0.003 to 0.55.

Nothing wins here, and that includes the competing method run on our own base. On the drifting base the backprop actor (0.655–0.668), CAST (0.655–0.659), both scoring variants (0.67–0.68) and DICE-RL at its own published settings (0.670–0.705) all sit within noise of the untuned base (0.671 ± 0.014). The point of the table is this column of near-zeros: it marks where our claim stops. DICE-RL’s +0.017 and CAST’s best variant at +0.006 are not distinguishable from each other at this seed spread: in the multitask setting neither rule has an edge. The flow-matching pipeline looks stronger in absolute terms (0.735–0.756), but its own base already scores 0.738 ± 0.022 , so it too extracts essentially nothing; its apparent lead is inherited from pretraining. Applying our final recipe — every fix and hyperparameter behind the +9.4 — changes nothing here (0.668 against base 0.671).

This bounds H2 rather than contradicting it. The update form converts critic signal into success; where there is no usable signal there is nothing to convert. It is also worth noting that every drift-base arm preserves the base within noise, including the backprop actor — consistent with a critic that is action-blind rather than action-exploitable. There is no steep hallucinated gradient here to ride. Finally, a methodological note: single-seed 20-episode endpoints ranged 0.63–0.78 across these same arms, and powered evaluation collapses that spread to the base value.

5.4 H3: the action-space constraint is what makes the form viable

Claim. Take the action-space constraint away, and the update destroys the policy.

Verdict. The factorial separates the two guards cleanly. Removing the base-restoring term costs everything: with the step clip alone, success is 0.000 on all five tasks, identical to using no constraint at all. Removing the step clip instead is survivable but erratic, swinging from +0.237 (red mug) to −0.223 (ketchup) against each task’s base. Only the pair clears base on every task. Table 7 then shows the shape of the constraint matters more than the machinery enforcing it — including a loss-space hinge that beats our anchor on two of three tasks. Two square cells of that table (always-on pull, hard projection) are withdrawn rather than reported: an audit found those runs used a different reward-field setting than the column’s other rows; their LIBERO columns are unaffected. The hinge’s square value is a single-seed plateau and remains provisional.

Table 5: The constraint factorial over five LIBERO tasks: clip and anchor switched independently, three seeds per cell. Without the anchor every run lands at 0.000; only the full pair clears base on all five tasks (see text).

clip	anchor	configuration	red mug onto plate (0.573)	ketchup into drawer (0.610)	bowl into top drawer (0.818)	juice into basket (0.785)	book into right caddy (0.578)	mean
×	×	unconstrained transport	0.000	0.000	0.000	0.000	0.000	0.000
✓	×	step containment only	0.000	0.000	0.000	0.000	0.000	0.000
×	✓	base-restoring only	0.810	0.387	0.807	0.860	0.423	0.657
✓	✓	full CAST	0.600	0.627	0.840	0.820	0.630	0.703

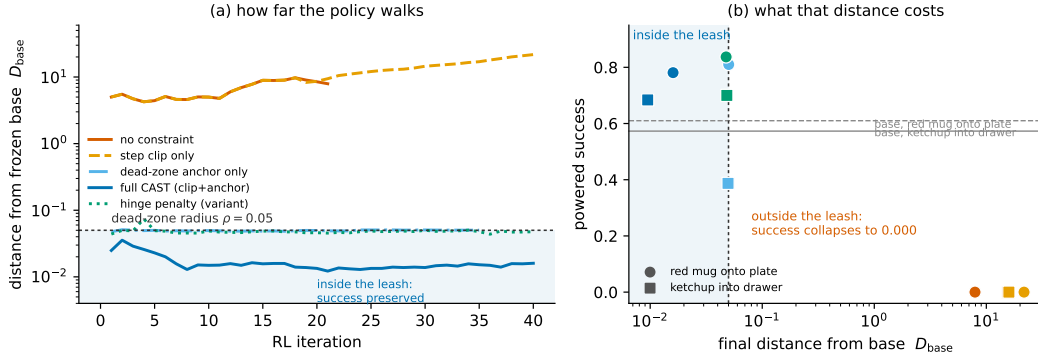


Figure 5: Distance from the base predicts collapse. (a) How far the policy walks from its frozen base during fine-tuning, $D_{\text{base}} = \mathbb{E} \|r_{\theta}\|_{\text{rms}}$, read from the training logs of the Table 5 arms on the red-mug task (log scale; curves end where the preemptible slot did). With no base-restoring term the policy leaves its base without bound, and the pointwise step clip does not prevent this — many individually-capped steps compound into a walk two orders of magnitude past the trust radius. (b) Why that matters: final distance against powered success, for both tasks. Every arm that ends inside the leash keeps or improves its base; every arm that ends outside scores exactly 0.000. This is the paper’s mechanism in one plot — the constraint is not a regulariser that trades a little performance for stability, it is the difference between a working policy and a dead one.

CAST bounds the target’s motion two ways: a pointwise clip on each particle’s displacement, and the dead-zone anchor. We claim neither over the other. The hypothesis is that some action-space constraint is necessary, so the experiment is the full factorial.

With no constraint at all, success is exactly 0.000 on both tasks, from bases of 0.573 and 0.610. The unconstrained action-gradient update destroys the policy. This reproduces inside our own system — our critic, our data, our budget — the behaviour DIPO and QSM show in their authors’ code at $125\times$ the interaction. That is the paper’s central negative result, and it is why we claim the constraint rather than the transport.

Figure 5 is where the mechanism becomes visible instead of inferred. Endpoint success tells us that constrained arms work; the residual trajectory tells us whether the policy was held near its base throughout, or merely happened to end near it.

The constraint’s shape is not critical; its presence is. Table 9 sweeps the dead-zone radius over two orders of magnitude. Every interior setting lands between 0.771 and 0.863 on the red-mug task and between 0.671 and 0.734 on the ketchup task — a spread no larger than this benchmark’s seed-to-seed variation.

The two endpoints behave differently, and the difference is the point. At $\rho=0$ the anchor becomes an always-on behaviour-cloning penalty and performance falls to roughly base level on both tasks: an always-on pull erases the residual. At $\rho\rightarrow\infty$ the dead-zone term goes

Table 6: The core comparison: full CAST against two simpler constraints, from one shared frozen base per task, matched critic, replay, budget and evaluation. Three training seeds each; square is 300-episode success, LIBERO is powered evaluation. °below the matched budget.

method	robomimic square	LIBERO red mug	LIBERO ketchup
DICE + dead-zone hinge (loss)	0.28	0.77	0.65
hard projection (target)	0.96	0.93	0.54
full CAST	0.94	0.78	0.68

inert, but this is not an unconstrained run — the weak pointwise restoring component and the displacement clip both remain, so it is a re-shaped constraint rather than an absent one. The genuinely unconstrained configuration is the first row of Table 5.

So a constraint must exist, and it must not be always-on. Between those two failure modes the setting is not critical, and no radius is singled out: $\rho=0.05$ sits inside a wide flat region, not at an optimum. The $\rho \rightarrow \infty$ cell is worth reading alongside Table 5, because together they say the same thing twice: that configuration still scores 0.803 because a weak pointwise restoring term survives in it, and when every restoring term is removed and only the displacement clip is left, success is 0.000. What the sweep varies is the shape of the tether; what Table 5 removes is the tether itself.

5.5 Does a simpler constraint suffice? CAST vs. two alternatives

Claim. CAST’s restoring field earns its place by being the most consistent constraint across tasks, not by dominating every one. Two simpler alternatives each match or beat it on some tasks but collapse on another: the loss-space hinge fails on square (0.29), and the hard projection fails on ketchup (0.54), while CAST is the only variant strong on all three. The lesson is that the constraint must act in action-target space — both target-space variants (projection and CAST) beat the loss-space hinge on square by a wide margin — and that a restoring field is the more reliable target-space instantiation than a hard projection. Square is three seeds at matched budget; the hinge’s seeds span 0.03–0.77, an instability the mean understates.

5.6 H4: the constraint has to act on the action target

Claim. The constraint has to act on the action target. The same shape placed in the loss, or on the parameter gradient, does not work.

Verdict. Two rows support it. The decisive row — giving the backpropagated actor our exact dead-zone shape inside its loss — is not yet run. We name it below as the experiment that would overturn this claim.

The most economical objection to this paper is that CAST is DICE-RL with a differently shaped regularizer, and that putting the constraint on the action target rather than in the loss is presentation rather than mechanism. Table 7 is built to test exactly that.

The table now separates where the constraint acts from what shape it takes, and only the first turns out to matter. Both action-target rows work and both loss-space rows fail: a hard projection onto $\|r\| \leq \rho$ reaches 0.790 and 0.733, our dead-zone anchor 0.781 and 0.684, while the always-on quadratic pull collapses to 0.549 and 0.601. We report the projection result even though it is the better of the two — on the ketchup task by 4.9 points, beyond that task’s seed spread. A simple hard cap on the residual is at least as good as the restoring field we build, which means our evidence supports the general claim (bound the displacement in action space) more strongly than it supports our particular instrument for doing so. That cap is also what ResFiT’s residual scale already implements, so the action-space constraint is not itself novel; what remains ours is the target-displacement form of the update, isolated in Table 3.

Table 7: Component ablations: replacing the reward field (top) or the restoring mechanism (bottom), holding the other at CAST’s setting. *quoted from Table 3; †single-seed plateau, provisional. “—” withdrawn or unmeasured (see text).

variant	acts on	square (3 seeds)	red mug onto plate (0.573)	ketchup into drawer (0.610)
(a) the reward field V_Q — constraint held at CAST’s				
gradient (CAST)	—	0.912–0.930	0.600	0.627
top- k (ranking only)	—	0.525–0.562	0.577	0.341
tilted ($\exp(Q/\tau)$)	—	0.560–0.584	0.697	0.378
(b) the restoring field — reward held at CAST’s gradient				
L_2 penalty + grad clip (DICE-RL)	loss + params	0.861–0.909*	0.550	0.587
always-on pull ($\rho=0$)	action target	running	0.549	0.601
hinge ($\ \tau\ - \rho$) $^2_+$	loss	0.991†	0.837	0.700
hard projection $\Pi_{\ \tau\ \leq\rho}$	action target	running	0.790	0.733
dead-zone anchor (CAST)	action target	0.912–0.930*	0.600	0.627

The first row is DICE-RL exactly as published, and it is worth being precise about what it already contains: a behaviour-cloning penalty in the loss and gradient-norm clipping at 1.0 on every actor step of every run in this paper. It constrains in both of the two places we say are wrong, and it still reaches only 0.550 and 0.587 against bases of 0.573 and 0.610.

The $\rho=0$ row makes the same point from inside our own method, and the correspondence is exact rather than analogous: by Proposition 1, CAST at $\rho=0$ with the clips removed is the DICE-RL residual actor’s update. Give the constraint the always-on quadratic shape that prior work uses and the policy returns to base level. This is the quantitative form of the toy’s finding that a penalty inside a loss must out-shout an unbounded gradient, and loses.

The sharpest objection is now measured, and it goes against the update form. DICE + hinge gives the backpropagated actor our dead-zone geometry inside its loss: same base, same critic, same $\beta_{BC}=100$, only actor_mode differs. We pre-registered the reading — if it matches CAST, the geometry is the whole story and the form is presentation. It does not merely match. It reaches 0.837 and 0.700 against full CAST’s 0.781 and 0.684 (and 0.627 on the matched arm we could verify), with its three evaluation seeds at 0.84/0.81/0.86 on the red-mug task. By our own criterion, on this benchmark the dead-zone geometry is doing the work, and the target-displacement form is not what earns the gain.

This does not erase Table 3 and Table 4, where the backpropagated actor without that geometry manages +0.9 against CAST’s +9.4. Read together, the three tables say something narrower than we set out to claim: a dead zone — a bounded region around the frozen policy with a free interior — is the ingredient that matters, and it can be delivered through a loss hinge, a hard projection, or our restoring field roughly equally well. Every configuration carrying that geometry lands in 0.68–0.84; both configurations without it (the L_2 penalty with gradient clipping, and the always-on pull at $\rho=0$) sit at base level or below, 0.549–0.601. What the target-displacement form buys is not a better ceiling but a way to get the geometry without differentiating through the critic at all — which is what makes it portable to the drop-in setting of Table 2, where the host method’s loss is not ours to edit.

5.7 How much of the critic should the update consume?

Claim. The critic’s gradient direction matters. CAST is not winning just by moving less.

Verdict. Holds on both benchmarks and both tasks.

Because reward enters only as a field, the critic can be consumed at lower resolution: transport toward its top- k ranked particles, or toward the cloud’s own $\exp(Q/\tau)$ -tilted density. Both delete the gradient direction and change nothing else.

The answer is the same wherever we ask it. On robomimic, where the critic generalizes, the value-only fields preserve can (0.91–0.95) but collapse on square, despite verified-healthy residuals and longer training. On LIBERO the same two substitutions fall to 0.577 and

0.697 on the red-mug task, and to 0.341 and 0.378 on the ketchup task — not merely below the full recipe but far below the untuned base, deficits of 34 and 31 points.

Two benchmarks, two bases, one conclusion: the ranking information a critic can supply is not a substitute for its gradient. CAST keeps the critic’s most informative signal and bounds only how far that signal may carry the policy in one step. Conservatism alone would not produce these numbers — it would produce the base.

5.8 Generality across generator classes

Claim. The update needs only a frozen noise-to-action map, so it transfers to other generator types.

Verdict. Holds on the four flow-matching tasks measured so far. Four remain, so we withhold a row mean.

On its own, stronger flow-matching base the published pipeline extracts essentially nothing per task: FM + DICE-RL scores -0.8 , -0.2 , $+2.2$ and -2.8 points relative to that base on the ketchup task, the orange-juice task, the red-mug task and the right-caddy task (mean -0.4) — the same near-zero extraction its actor shows on the drifting base ($+0.9$). Swapping only the actor update helps on three of the four: CAST on the identical base and budget beats FM + DICE-RL by $+5.6$ (ketchup task), $+6.9$ (red-mug task) and $+1.6$ (right-caddy task), reaching 0.738 and 0.878 on the first two, which is $+4.8$ and $+9.1$ over that base.

The exception is instructive. On the orange-juice task the flow-matching base already scores 0.917 and neither update improves it (-1.8 for CAST, -0.2 for DICE-RL). More generally, the never-below-base behaviour CAST shows on all eight drifting-base tasks does not carry over here: on the orange-juice task and the right-caddy task it ends 2.0 and 1.2 points below where it started. Fine-tuning a base that is already near its ceiling is a different problem from lifting a weak one, and our trust region is calibrated for the latter.

5.9 Scope: when is the constraint needed, and does it transfer?

The mechanism sections establish that the constraint is necessary in our setting. Three scoping questions then determine how far the claim reaches, and we test each with update rules already validated in this paper — the field-target update and the backpropagated DICE-RL actor — rather than with methods that fail everywhere and so can distinguish nothing.

Q1: is the failure specific to fine-tuning from narrow data? The collapse mechanism needs a critic whose competence is a thin tube around demonstrated behaviour. We therefore cross the same update rules over data regimes: narrow-coverage fine-tuning (LIBERO, robomimic; measured throughout this paper) against a broad-coverage base — a DMC cartpole base distilled deliberately from multiple noisy SAC checkpoints, so the critic’s training tube is wide. The residual question inside Q1 is already settled by Table 2: DIPO, QSM and DQL fine-tune the full policy with no residual head and collapse regardless, so the residual parameterization is packaging, not disease.

Q2: does the failure need sparse reward? The same DMC base supports a controlled pair — swingup (dense) and swingup_sparse — identical dynamics, identical base, identical critic architecture, only the reward signal differing. Table 8 runs three arms on each. The registered prediction, from the degenerate-fixed-point analysis of Section 5.2, was that dense reward lets a drifted policy recover while sparse reward does not, and the completed runs bear it out in the direction that matters: under dense reward the unconstrained update not only survives but ends above its base in interim evaluations (~ 540 against a base of ~ 430), the exact opposite of its sparse-manipulation behaviour. The correct reading of sparsity is therefore not that it causes the drift — the drift is the update’s doing — but that it removes the road back: once off-manifold with no successes, every reward is zero and the critic has nothing with which to guide a return. Dense reward keeps that road open. This also resolves an apparent paradox in our own results: our sparse-reward benchmarks are

Table 8: DMC: three actor updates fine-tuning the same frozen distilled base, dense vs. sparse reward. Protocol follows standard DMC reporting (Tassa et al., 2018): return over fixed 1000-step episodes (bounded at 1000 by construction), mean of 10 evaluation episodes, quoted at a matched 400K environment steps. Swingup averages two training seeds; second seeds for the other rows are running. [◦]below the matched budget; [◦]high eval variance; ^ttuned reward step (see text).

task	base	unconstrained	backprop $-Q$	CAST
dense reward				
cartpole swingup	429	575	318	715
cartpole balance	378	815 [◦]	844	911
quadruped run	238	97 [◦]	283	312 ^{◦t}
finger spin	3	0 [◦]	187 [◦]	187 [◦]
sparse reward				
cartpole balance (sparse)	90	880 [◦]	843 ^v	1000
ball-in-cup catch (sparse)	86	940 [◦]	66 [◦]	925

survivable for constrained methods precisely because the constraint prevents the one-way step through the trapdoor that sparsity cannot undo.

The sparse-swingup variant is excluded from the table: its base is too weak under sparse reward to support any conclusion, as reported previously, so the sparse test is carried by the balance pair.

The tuned quadruped arm. The shipped DMC recipe runs $\eta_Q=0.5$, the setting our own sweep rates worst; at $\eta_Q=1$ CAST reaches 312 where the untuned arm reads 267 (a dead-zone variant reads 301), against backprop’s 283 at nearly $3\times$ the budget. The tuned value is reported with this disclosure.

Excluded tasks. Walker-run and humanoid-walk are excluded: their one-step distills retain almost none of the teacher (17 from a 691 teacher; 1 from 468), and a $4\times$ -capacity, $4\times$ -epoch distill plateaus at the same loss — distillation from 100 demonstrations, not fine-tuning, is where the pipeline breaks on these morphologies, so no fine-tuner comparison is measurable there. Humanoid-run and manipulator bring-ball are excluded for failed teachers (returns 10 and 0.2).

Tuning. The quadruped CAST cell uses $\eta_Q=1$ rather than the shipped 0.5: both tuned variants (stronger reward step, and a dead-zone at $\rho=0.05$ rms) beat the untuned arm (312 and 301 against 267) and the backpropagated actor (283) at less than half its budget. Every other CAST cell runs the recipe below unchanged.

How CAST is configured here. We do not re-tune for DMC. The arm runs the robomimic recipe unchanged: actor_mode field-pointwise, V_Q from the critic gradient with reward step $\eta_Q=0.5$, anchor step 0.05, total displacement clip $\delta_{\text{tot}}=0.15$, restoring step 1.0, $K=16$ noise samples per state, 3-step returns, and a one-step (flow_steps=1) drifting base distilled from the SAC checkpoints. The dead-zone radius is $\rho=0$ here, i.e. the restoring term is always on rather than gated — the setting Table 9 shows is worst on LIBERO. The DMC numbers are therefore an untuned transfer, not a best case, and should be read as a scope test rather than a performance claim.

Q3: does the bound transfer to other algorithms? Two transfers are tested, both on validated hosts. First, the dead-zone geometry inside the backpropagated actor’s loss — the DICE + hinge row of Table 7 — asks whether our constraint helps the strongest existing update when expressed in its native form. Second, as an external probe, DIPO’s own action-improvement loop is given a per-round cap on total displacement and a consistent buffer (two config-gated switches in its official code, byte-identical when off); DIPO is the weakest member of the family (+0.004 in Table 2), so any recovery is attributable to the bound alone. Both runs are in progress and will be reported whichever way they land.

Table 9: Recipe sensitivity on LIBERO: each row changes one component; Δ is against the full recipe. Removing any component costs up to 23 points, the critical one is task-dependent, and the dead-zone radius is flat across its interior with failures only at the endpoints (see text).

	red mug onto plate (base 0.573)		ketchup into drawer (base 0.610)	
	success	Δ vs. full	success	Δ vs. full
full recipe	0.781		0.684	
remove or weaken one component:				
anchor measures each dimension separately $\rightarrow$ one vector norm	0.662	-0.119	0.655	-0.029
dead zone removed ($\rho = 0$, pull always on)	0.549	-0.232	0.601	-0.083
reward step halved ($\eta_Q: 1 \rightarrow 0.5$)	0.769	-0.012	0.615	-0.069
multi-step returns removed ($n: 3 \rightarrow 1$)	0.782	+0.001	0.512	-0.172
loosen one component beyond the shared recipe:				
wider trust region ($\delta_{\text{tot}}: 0.15 \rightarrow 0.25$)	0.874	+0.093	0.689	+0.005
stronger reward step ($\eta_Q: 1 \rightarrow 2$)	0.802	+0.021	0.729	+0.045
dead-zone radius sweep:				
$\rho = 0$ (always-on penalty)	0.549	-0.232	0.601	-0.083
$\rho = 0.02$	0.863	+0.082	0.734	+0.050
$\rho = 0.05$ (recipe)	0.781		0.684	
$\rho = 0.10$	0.812	+0.031	0.694	+0.010
$\rho = 0.20$	0.771	-0.010	0.671	-0.013
$\rho = \infty$ (dead zone inert)	0.803	+0.022	0.671	-0.013

5.10 Robustness, and what the bound actually does

The remaining knobs are task-dependent, and they track the critic rather than the task’s difficulty. On the ketchup task, whose critic carries a strong action-gradient ($\|\nabla_a Q\| \approx 0.55$), 3-step returns are worth 17 points and a doubled reward step 4.5, while widening the movement bound alone does nothing: this task wants more reward signal, not more room to move. On the red-mug task, whose gradient is an order of magnitude weaker (≈ 0.035), the pattern inverts — returns and step size are within noise, but widening the trust region is worth +9.3 points. We report this as a regime difference, not a tuned optimum: it is one seed, and the red-mug task carries a ± 9 -point spread, so the direction is trustworthy and the magnitude is not.

Is the gap just a better step size? We tested this directly on the red-mug task by sweeping each method’s own step-size knob under the identical protocol. The backpropagated actor’s learning rate over a $10\times$ range gives 0.550, 0.548 and 0.578 — every setting at or below the untuned base (0.573). Our reward step over an $8\times$ range gives 0.775, 0.817, 0.908 and 0.750 for $\eta_Q \in \{1, 2, 4, 8\}$ — every setting 18–33 points above that same base. The two bands do not overlap at any step size either method was given. A larger step does not turn the backpropagated update into the field update. (Both sweeps are single-seed, so the ordering within each band should not be read as a tuned optimum; the band separation, at 18 points minimum, is well outside seed noise.)

How often does the bound actually bind? The clip only ever reduces a displacement, so whenever a particle’s proposed motion is already shorter than δ_{tot} the update is exactly the unbounded one. In the Section 3.3 setting at our operating point, the clip is active on 17.3% of sampled actions and the median proposed displacement is only $0.12 \delta_{\text{tot}}$. Tightening the cap to 0.05 raises the binding rate to 26.3%; loosening it to 1.0 drops it to 4.8%.

These numbers say what kind of object the trust region is. It is not a step-size controller: for most proposals the critic’s own magnitude still sets the distance, exactly as in the unconstrained update. What the constraint governs is the tail — the minority of proposals where an imperfect critic asks for a displacement far outside anything its data supports, and which, left alone, are enough to end the run (Table 5, first row). That is why our claim is about where the critic’s proposal is mediated, not about how far the policy travels on average. It is also why we do not adjudicate between the two implementations of the constraint: both truncate the same tail.

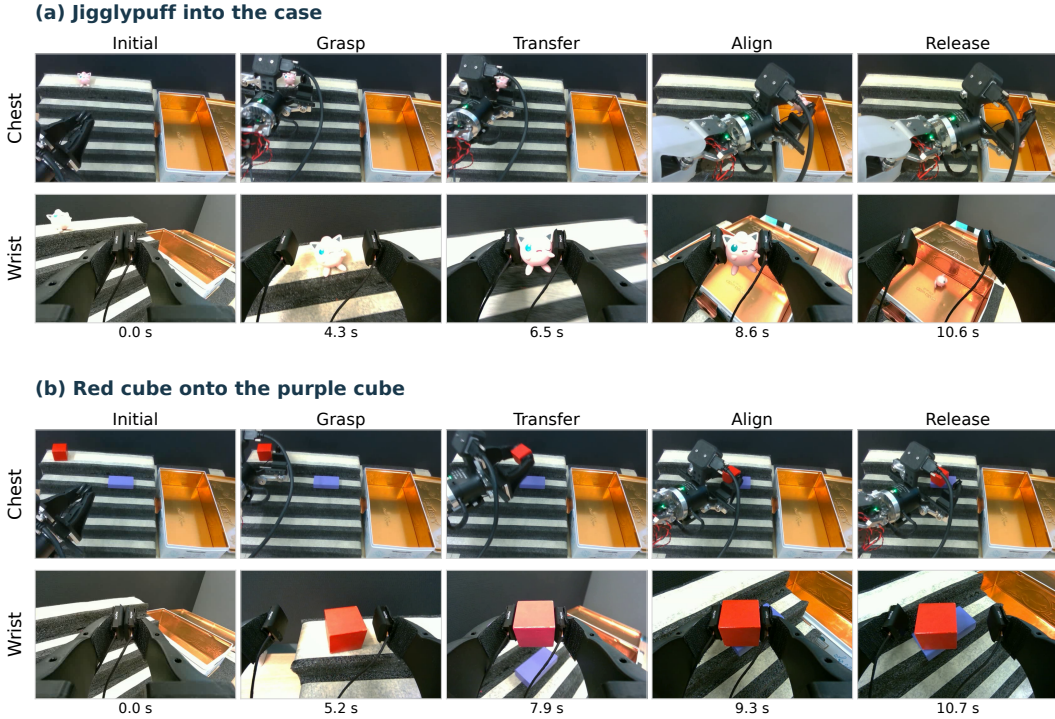


Figure 6: Two real-robot tasks on a hand-made staircase. Each task is shown as a five-stage sequence, with chest and wrist views in paired rows: initial arrangement, grasp, transfer, alignment, and release. The wrist view shows the toy released into the case in (a) and the red cube resting on the purple cube with the gripper open in (b). Frames are extracted from successful teleoperated demonstrations, not autonomous evaluations of the compared methods. Times are nominal video times at 15 fps; the two views are approximately aligned by frame index. A consistent top-background crop is applied to all panels.

5.11 Real-robot fine-tuning on stair-based manipulation tasks

We extend the study to physical manipulation on an OpenArm platform, asking whether post-training can improve a demonstration-trained policy when the feedback available after each attempt is a human success/failure label. The real-robot study is in progress: the task setup and demonstrations are available, while the four-way quantitative comparison below remains to be completed. This study tests refinement of an existing skill rather than learning a new policy from scratch.

Platform and tasks. The robot operates its left seven-joint arm and gripper using RGB images from a chest-mounted camera and a left-wrist camera, together with left-arm joint and gripper positions and a fixed task-language prompt. We consider two tasks: Jigglypuff into the case, in which the robot grasps the toy from a stair tread and releases it into an adjacent open case; and red-on-purple stacking, in which it picks up the red cube and places it on the purple cube. Both use a hand-made staircase. The stepped support surfaces introduce different approach heights and nearby edges, making grasping and placement sensitive to alignment; stacking additionally requires the upper cube to be positioned accurately before release. Figure 6 illustrates the tasks through successful teleoperated demonstrations from both views.

Pretrained policy and refinement methods. For each task, we first train a flow-matching BC policy on teleoperated demonstrations. The current policy predicts 30-step chunks of absolute left-arm joint and gripper targets at 15 Hz, using a 10-step Euler sampler. The

Table 10: Planned real-robot comparison. Task success rate for a shared pretrained FM checkpoint per task and three fine-tuning methods. All eight result entries are not yet measured; “—” is a placeholder, not zero success. Evaluation trial counts and uncertainty will accompany the completed measurements.

Method	Online refinement data	Jigglypuff into case	Red cube onto purple cube
Base FM	None	—	—
Success-only BC refinement	Successes only	—	—
DICE-RL	Successes and failures	—	—
DICE-RL + CAST (ours)	Successes and failures	—	—

deployed interface executes a prefix of each chunk before replanning. The comparison uses one fixed base checkpoint per task and four conditions:

- Base FM: deploy the demonstration-trained checkpoint without online refinement.
- Success-only BC refinement: initialize from that same checkpoint and continue supervised flow-matching training using only successful online rollout trajectories. Failed attempts are excluded from these imitation targets.
- DICE-RL: fine-tune a residual actor and critic over the same frozen base using both successful and failed online rollouts, with the standard DICE-RL actor objective (Sun & Song, 2026).
- DICE-RL + CAST: use the same residual RL setup and both rollout outcomes, replacing the actor update with CAST’s clipped, base-restoring action-target regression.

Thus all three refinement conditions start from pretrained weights; none trains a task policy from scratch. Success-only BC tests how much improvement can be obtained by imitating successful practice, while the two RL conditions test whether failed experience and the choice of actor update yield further benefit.

Rollout collection and evaluation protocol. The evaluation interface records camera observations, measured robot states, and executed commands during each attempt. After stopping the robot, an operator assigns success or failure and chooses whether to retain the trajectory. The label is the sparse task-outcome feedback; it is not inferred from the BC loss or a critic score. For the RL comparison, valid failed trials must be retained alongside successful ones, with discarded or invalid trials accounted for explicitly to avoid success-selection bias.

The planned comparison matches the starting checkpoint, robot-interaction budget, reset and object-placement distribution, and evaluation conditions. DICE-RL and DICE-RL + CAST additionally share the critic architecture, critic-training procedure, demonstration-replay settings, exploration, and update schedule, isolating the actor update. All methods use the same observation/action interface and execution limits. Replay transitions must correspond to the commands actually executed, including chunk-prefix execution and command limiting, rather than unexecuted predictions. Evaluation trials are held out from all refinement data.

We will count Jigglypuff placement as successful when the toy is released fully inside the case and remains there, and stacking as successful when the red cube remains supported by the purple cube after release and gripper withdrawal. A common post-release observation interval and matched trial counts will be fixed before scoring the four methods. Table 10 reserves the success-rate comparison; the demonstration sequences establish the task setup but do not supply any of its quantitative entries.

6 Conclusion

This paper is about one narrow, common regime: post-training a generative policy that was pretrained on a small set of demonstrations, online, under sparse binary reward. Our experiments locate the boundary of that regime from both sides. Inside it, any update that lets the critic displace actions without an action-space bound destroys the policy — in our system and in the official implementations of an entire method family — because the critic is only competent in the narrow tube its data covers and sparse reward offers no route back once the policy leaves it. Outside it, the claim weakens but does not vanish: under dense reward the failure stops being terminal, yet on three DMC tasks CAST still wins — dense reward removes the catastrophe, not the cost of wandering, and the unconstrained update still falls below its base on quadruped (Table 8). Only at saturating critic coverage (Figure 2a) does backpropagation become the better tool. Within the regime, the missing update rule is field-target regression under an action-space bound anchored to the frozen base. CAST expresses reward and anchoring as composable velocity fields over action particles, and recovers the backpropagated direction in the small-step limit. Its working component is the action-space trust region: the critic sets the direction, while a bound — a pointwise clip on each particle’s displacement, a restoring pull outside a radius of the frozen base, or both — sets how far any single update may carry the policy.

Given the same base, critic, data and budget, the field-target form extracts far more than backpropagating that critic: +55 points on robomimic square, and +9.4 versus +0.9 across the eight hardest LIBERO-90 tasks, with the same ordering after swapping in a stronger flow-matching base — so the effect belongs to the update rule, not the starting point. Against a critic-free policy gradient at matched budget, CAST is at parity while needing none of the noise injection that makes likelihood ratios definable for a one-step generator. The ablations then pin the mechanism: removing every action-space bound gives exactly 0.000 success on all five factorial tasks, the step clip alone also gives 0.000, the always-on penalty ($\rho=0$, exactly a BC term) falls back to base, and only configurations with a restoring term toward the frozen base survive. The constraint that matters is not a limit on how fast the policy moves but a tether to where it started.

Limitations. Three specific results run against our thesis and we state them plainly. On the Study-3 caddy task the backpropagated actor ends above CAST (0.653 versus 0.635); on two flow-matching tasks the un-finetuned base beats every fine-tuner including ours (0.917 on the orange-juice task, 0.620 on the right-caddy task); and on robomimic can the flow-matching pipeline edges us on two of three seeds. Single-seed variation is also substantial on LIBERO — repeating the frozen recipe on the red-mug task with a second seed moves the result by 8.9 points — so per-task deltas smaller than roughly ten points, including our η_Q and δ_{tot} sweeps, indicate direction rather than magnitude. The claims that survive this variation are the ones separated by margins far larger than it: the +9.4 versus +0.9 mean gap, the non-overlapping step-size bands, and the below-base collapse when the critic’s gradient is deleted.

The gains track how much a critic actually knows. When a single critic is shared across all eight LIBERO tasks its action-gradient collapses and no critic-based update — ours or the state of the art’s — improves on its base; when a base is already near its ceiling, CAST can end slightly below it (−2.0 and −1.2 on two flow-matching tasks), since our trust region is calibrated for lifting weak bases rather than for refining strong ones. Our critic-trust diagnosis is also post hoc: choosing the reward-field resolution online from critic-trust signals is the natural next step, as is testing whether the reward step should adapt per task. Quantitative fine-tuning results are currently limited to simulation; Section 5.11 describes the ongoing physical-robot study, whose four-way success-rate comparison remains to be completed.

We should be precise about which part of the story generalizes, because two claims are easy to conflate. The update rule is base-agnostic: it consumes a frozen noise-to-action map, a residual head and a critic, and never touches how the base draws its action, and it transfers unchanged to a flow-matching base, where dropping it into DICE-RL raises success on square from 0.831 to 0.934; the diffusion-base pair converges to a shared ≈ 0.96 ceiling with CAST

arriving $1.6\times$ sooner (Table 2). Its size of benefit, however, is not base-agnostic: on the flow-matching LIBERO base the same recipe is a wash against the base (0.756 vs 0.753) and only +1.3 points over DICE-RL across the seven tasks where both ran. What does not generalize is the economy of the construction. A drifting model is pretrained by field-target regression, so on that base the fine-tuner reuses machinery the policy class already has; a flow-matching model is pretrained by conditional velocity regression, so there the same update is bolted on rather than inherited (and costs one base forward pass per particle per integration step). The principle we can defend is therefore narrow: what keeps a critic-driven update on the data is a bound on the total displacement in action space, not a limit on step size. The claim stops at the location of the constraint. On the two LIBERO tasks a hinge penalty and a hard projection onto $\|r\| \leq \rho$ both beat our dead-zone anchor, while on square the anchor wins by a wide margin (Table 7); the anchor is one workable instrument rather than the necessary one, and which instrument is best appears task-dependent. The distinction is not cosmetic. A step limit constrains speed, and a drift assembled from permitted small steps still arrives; in our isolation the limit alone leaves the policy at zero success. A behaviour-cloning penalty restores, but does so everywhere, including the region where the critic is reliable, and must therefore trade far-field containment against near-field progress with a single coefficient; it also fails alone. The dead zone separates those jobs.

We deliberately claim no more than this. We have no principled rule for choosing ρ — it is tuned, not derived — and the argument for why the dead zone should exist is an argument about matching the regularizer’s spatial profile to the critic’s reliability, not a bound. Deriving ρ from a measure of local critic reliability is the obvious next question, and we leave it open.

Reproducibility statement

We release the training framework, the implant that runs our update inside the official DICE-RL harness, and the exact configuration for every arm, including the family-shared recipe and all seeds. Evaluation protocols are stated with each table: robomimic scores are 300-episode evaluations, LIBERO scores use powered evaluation (100 rollouts $\times 3$ evaluation seeds), and every reported number is an average over the stated seeds, never a best-of. Each run fits on a single L40S or A100 GPU (8–20 hours). The toy experiments ship as standalone scripts that regenerate Figure 1 and Figure 2 from scratch.

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A The kernel transport operator

Given query particles x , attractors p^+ (weights w^+), and repulsors p^- , all distances $d(x, p)$ are first divided by a global scale — the weighted mean pairwise distance — so the operator is invariant to the action scale. For each kernel radius $R \in \{0.02, 0.05, 0.2\}$ a soft mutual-matching affinity

$$A = \sqrt{\text{softmax}_{\text{row}}(-d/R) \odot \text{softmax}_{\text{col}}(-d/R)} , \quad (10)$$

weighted by w , converts matched targets into forces: each matched attractor contributes a pull scaled by the query’s total repulsor affinity, each matched repulsor a push scaled by its total attractor affinity, so a query contested equally from both sides feels no net force. When a target block is the query set itself, self-pairs are excluded symmetrically — masking only one block leaves uncanceled self-attraction, producing a spurious field even when the two distributions coincide. Forces are summed over radii; per-radius normalization by the root-mean-square force is enabled for the restoring field (giving a constant-strength pull) and disabled for reward fields (preserving magnitude information). The implementation is released with the code.

B Toy two-regime construction

Both regimes share one recipe: a frozen base MLP maps d -dimensional noise to d -dimensional actions, a critic is regressed onto $(a, r(a))$ pairs drawn from the base’s own samples, and each fine-tuner runs against that frozen critic with the same budget. The easy regime uses $d=2$ with 200 critic points; the hard regime uses $d=16$ with 60 — coverage, not dimension alone, is the operative variable, and Figure 1 crosses the same boundary by varying data at fixed dimension. Arms: backpropagating $-Q$ with and without a BC anchor ($\lambda=1$), and the bounded field update with tilted and top- k scoring. Three seeds; curves in Figure 2 are seed means. The script regenerates the figure from scratch.

B.1 Reward-step sweep

CAST’s reward step η_Q is the one hyperparameter we varied. We report the whole sweep here because the per-task optima disagree, and because the main tables deliberately do not use them: every CAST number in Table 4 runs a single shared η_Q across all eight tasks, the same courtesy the baselines get by being reported at their published recipes. Tuning per task would raise our numbers — 0.880 instead of 0.802 on the red-mug task — and would not be a like-for-like comparison.

Table 11: Reward-step sweep, powered evaluation (last-3 checkpoints, single training seed). The optimum is task-dependent and non-monotonic: $\eta_Q=4$ is best on the red-mug task, $\eta_Q=1$ on ketchup, and $\eta_Q=8$ is worst on both. $\eta_Q=2$ is best on neither individually but has the best mean across the eight tasks of Table 4, which is why it is the setting we ship. On the flow-matching base the ordering differs again — $\eta_Q=2$ beats $\eta_Q=1$ on all three tasks measured — so the shared setting is a compromise across bases as well as tasks. The completed flow-matching row confirms both halves of the drifting story: the optimum is task-dependent ($\eta_Q=4$ jumps the red-mug task to 0.950 while barely moving ketchup), and pushing too far collapses ($\eta_Q=8$ drops the caddy task to 0.053, far below its 0.683 base).

	base	task (own base)	$\eta_Q=1$	$\eta_Q=2$	$\eta_Q=4$	$\eta_Q=8$
	drifting	red mug onto plate (0.573)	0.775	0.802	0.880	0.768
	drifting	ketchup into drawer (0.610)	0.737	0.720	0.721	0.686
	flow matching	ketchup into drawer (0.690)	0.738	0.773	0.727	0.747
	flow matching	red mug onto plate (0.787)	0.878	0.887	0.950	0.853
	flow matching	book into front caddy (0.683)	0.667	0.783	0.673	0.053

The flow-matching rows also correct an earlier reading of ours. At $\eta_Q=1$ CAST sits below the base on the front-caddy task (0.667 against 0.683) and gains only +0.003 on the eight-task mean; at $\eta_Q=2$ the same task gains +0.100. The flow-matching column was therefore under-tuned rather than resistant to the method, and we had previously attributed its flat result to limited headroom — an explanation the per-task numbers do not support, since gains are the same (+0.003 mean) for tasks whose base is below 0.72 and above it.

B.2 Measured diagnostics behind the mechanism claims

Three claims in the main text rest on direct measurements rather than on outcomes. We give the numbers here so they can be checked.

(i) How weak the shared critic’s action signal is. Table 15 reports that no update rule moves the shared-critic base under a single noise draw. The measurement below shows why a gradient-following update struggles there: the shared critic barely changes its value when the action moves, so the direction a method would climb is dominated by noise. We flag one limit of this diagnostic. It also reports a narrow spread across the K candidates, which would predict that selection fails too — and Table 16 shows it does not: best-of-16 selection recovers 5.4 points for DICE-RL and 2.3 for CAST. A spread of 0.7% is evidently still enough to rank candidates usefully, so this table explains the weakness of the gradient channel, not an absence of signal overall.

(ii) Why QSM collapses on the first update. QSM’s actor loss retrain the denoiser to emit $\eta \nabla_a Q$ in place of the noise it was pretrained to predict, with no behaviour-cloning term. Whether that is survivable depends on whether the two are on the same scale. They are not: measured on the pretrained square base with a freshly initialised critic, $\|\nabla_a Q\| = 0.0127$ against $\|\epsilon\| = 5.212$. At QSM’s published coefficient the target is 2.4% of what the denoiser currently outputs, so the pretrained map is overwritten rather than adjusted — consistent with the observed $0.284 \rightarrow 0.000$ within a single evaluation interval. No coefficient resolves this: reaching parity needs $\eta \approx 400$, at which point the denoiser is copying a random critic. (scripts/diag_qsm_scale.py reproduces this in seconds.)

Table 12: Shared-critic action-sensitivity on the LIBERO hard-8 multitask policy, measured on $B=8$ states $\times K=16$ candidates. The two rows are the extremes across tasks; the diagnostic labels the first “alive” and the second “DEAD (flat critic)”. Even where the gradient has magnitude, a 0.05 rms action step changes Q by under 1 part in 40 of $|Q|$, and the 16 candidates are scored within 0.7% of one another.

	gradient “alive”	gradient “dead”
$\ \nabla_a Q\ $ (per-particle mean)	5.49×10^{-1}	3.10×10^{-3}
$ \Delta Q $ for a 0.05 rms action step	9.21×10^{-4}	7.05×10^{-5}
$ Q $ mean	0.043	0.579
$\text{std}(Q)/ Q $ across $K=16$	0.0418	0.0003
$Q_{\max} - Q_{\min}$ across $K=16$	6.94×10^{-3}	7.85×10^{-4}

Table 13: Scale of QSM’s score-matching target against the denoiser’s own output, square base, critic at initialisation.

η (q_grad_coeff)	1	10 (published)	100
$\ \eta \nabla_a Q\ $	0.013	0.127	1.266
ratio to $\ \epsilon\ $	0.0024	0.0243	0.2428

(iii) How far each constraint actually lets the policy travel. Figure 5 plots this; the end-points are given here. The trust radius is $\rho = 0.05$. Note that the step clip alone permits the largest excursion of all — many individually capped steps compound — and that full CAST stops at roughly a third of its own allowance, which is what the loosening sweep in Table 11 probes.

Table 14: Distance from the frozen base at end of training, $D_{\text{base}} = \mathbb{E}\|r_\theta\|_{\text{rms}}$, red-mug task, $\rho = 0.05$.

configuration	D_{base}	powered success
no constraint	7.92	0.000
step clip only	21.70	0.000
dead-zone anchor only	0.050	0.810
hinge in the loss	0.048	0.837
full CAST	0.016	0.600

B.3 The multitask setting

Table 4 fine-tunes each task with its own residual head and critic, routed by task ID at deployment — the architecture published LIBERO RL fine-tuners use. The alternative is one policy and one critic trained across all eight tasks jointly. Under the single-noise-draw evaluation used throughout this paper, no arm moves its base by more than seed noise (Table 15), and Table 12 supplies a mechanism: 16 candidate actions scored within 0.7% of one another, and a 0.05 action step moving Q by under one part in forty.

That reading turns out to be an artefact of the evaluation, and we correct it here. A single noise draw reproduces the base whenever the learned residual is near zero, so it cannot distinguish “the update rule did nothing” from “the update rule learned something the evaluation never samples.” Re-scoring the same checkpoints under the best-of- N critic selection that DICE-RL’s own LIBERO evaluation uses ($N=16$, max_q_min) separates the two, and the picture changes (Table 16): DICE-RL gains 5.4 points over the shared-critic base and CAST gains 2.3, so the shared critic does carry usable signal — it is only unusable when a single draw has to find it. We report this plainly because it cuts against us: in the multitask setting CAST improves on its base but trails DICE-RL by 3.0 points, reversing the per-task ordering of Table 4. Our displacement bound is what keeps the residual close to the base, and when the critic is this weakly action-sensitive that conservatism costs more

than it protects. We take this as the honest boundary of the method rather than a tuning deficit.

Table 15: The bound on H2: where the critic has no usable signal, no update rule helps — including ours. LIBERO hard-8 with a single policy and a single critic trained on all eight tasks jointly, in contrast to the per-task routing of Table 4. Powered evaluation (100 rollouts $\times$ 3 evaluation seeds; one column per training seed; 100 RL iterations). This is a negative result and we report it as one: read the Δ column, where nothing moves its own base by more than seed noise (± 0.014). Two rows use the backpropagated actor and they are not duplicates — “its published recipe” is DICE-RL as shipped ($\beta_{BC}=100$, best-of- N exploration), while “our recipe” is the same actor under the settings CAST uses ($\beta_{BC}=10$), so it is the matched control. Δ is against each row’s own base, the only fair comparison across the two blocks: the flow-matching numbers are larger in absolute terms purely because that base starts higher. The final block is an ablation of one component — how the critic scores the K candidates before displacement — and is not a separate method; the corresponding single-task sweep is Table 7.

method	s10000	s10001	s10002	Δ vs. own base
drifting base				
drifting base (no RL)	0.671 ± 0.014			—
drifting + DICE-RL (its published recipe)	0.705	0.690	0.670	+0.017
drifting + backprop actor (our recipe)	0.657	0.655	0.668	−0.011
drifting + DICE-RL + CAST (our recipe)	0.655	0.668	0.659	−0.010
flow-matching base				
flow-matching base (no RL)	0.738 ± 0.022			—
flow matching + DICE-RL	0.756	0.735	0.737	+0.005
flow matching + DICE-RL + CAST	0.746	0.729	0.746	+0.002
ablation: how CAST scores candidates (drifting base)				
top- k scoring (mean Q of best k)	0.665	0.684	0.668	+0.001
tilted scoring (Q -weighted average)	0.677	0.627	0.672	−0.013

Table 16: Under the selection protocol, the multitask setting is not inert — and CAST trails DICE-RL. The same shared-critic checkpoints as Table 15, re-scored with best-of- N critic selection ($N=16$, $\max_q \min$), which is the protocol DICE-RL’s own LIBERO evaluation uses; Table 15 instead draws a single noise sample, under which a near-zero residual is indistinguishable from its base. Drifting base, LIBERO hard-8, 100 rollouts $\times$ 3 evaluation seeds. Read together with Table 15: the ranking is not preserved across protocols, so the multitask conclusion depends on how candidates are sampled at deployment.

method	best-of-16	Δ vs. base	single draw (Table 15)
drifting base (no RL)	0.669	—	0.671
drifting + DICE-RL	0.722	+0.054	0.688
drifting + DICE-RL + CAST	0.692	+0.023	0.663

C When CAST wins, and when it must lose

Every result in this paper, wins and losses alike, is predicted by one property of the update rule, and we state it here explicitly.

CAST is a fixed trust region anchored at a frozen reference. The anchor gives the residual an equilibrium: the clipped reward field pushes outward with at most η_Q per step, the restoring field pulls back toward the frozen base, and the balance point sits near $\|r\| \approx \rho + \delta_Q/\eta_{anc}$. The consequence is structural: CAST can only ever express the best policy that lies inside a small action-space ball around π_0 . The same property that prevents the critic from dragging the policy off the data manifold prevents it from dragging the policy anywhere far — including toward a genuinely better policy outside the ball. This is

measured, not hypothetical: on transport the residual settles at $\|r\| \approx 0.007$ while the critic requests displacements of ~ 0.25 per step (Table 3), and in the flagship LIBERO regime the residual sits an order of magnitude inside the dead zone (Figure 1d) — there the bound is slack and CAST wins on direction quality alone.

Two conditions decide the outcome. (i) Does a substantially better policy lie inside the ball — does the base need refinement rather than new behaviour? (ii) Is the critic dangerous outside the ball — few demonstrations and sparse reward, so that off-manifold Q is extrapolation? The four cells of this 2×2 are each occupied by one of our measurements:

	critic dangerous outside	critic trustworthy outside
better policy inside the ball	CAST dominates: LIBERO hard-8 (+9.4 vs +0.9), square on every base, can, DMC swingup	CAST ties at the ceiling but arrives at less compute: square, retrained base (0.955 vs 0.941 at 352K; both ≈ 0.96 converged)
better policy outside the ball	nothing works: multitask LIBERO with a shared, action-blind critic (Table 15)	CAST loses: transport (0.594 vs 0.912) — the base barely solves the task, and a critic trained on abundant successful experience earns the trust that the bound withholds

Transport occupies the one cell where a bounded update must lose. The backpropagated actor wins there through the exact mechanism this paper documents as a flaw: its loss-space penalty leaks — the regularizer competes against an unbounded gradient and loses — so its residual can grow as large as the task demands. Where the critic deserves that trust, the leak is an advantage; where it does not, it is the collapse of Table 2.

Scope, restated. CAST is the right tool when a competent base cloned from few demonstrations must improve under sparse reward with a per-task critic — the post-training setting of Section 3.1. It is the wrong tool when the base lacks the skill outright and the critic has earned trust far from the data; there, methods that follow the critic freely are correct to do so. The practical design rule this suggests — grow the radius with the critic’s earned trust — is tested by a loosened-leash transport run in progress at the time of writing.